RETROFITTING OF A MANUAL WHEELCHAIR INTO AN ADAPTIVE VOICE CONTROLLED HYBRID MOBILITY SYSTEM FOR PAEDIATRIC MOTOR-IMPAIRED USERS

BY

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(2020/9415)

BELLS UNIVERSITY OF TECHNOLOGY, OTTA

JULY 2025

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A Project Submitted to the in Partial Fulfilment of the Requirements for the Award of bachelor's degree of Engineering (B. Eng.) in Mechatronics Engineering at the Department of Mechatronics Engineering, College of Engineering, Bells University of Technology, Ota.

JULY 2025

DECLARATION

I hereby declare that this research work was conducted under the supervision of Engr. Daniel Akinwande in the Department of Mechatronics Engineering, College of Engineering, Bells University of Technology, Ota, Nigeria. I confirm that this dissertation has not been submitted, either in whole or in part, for the award of any degree at any other institution. All sources of data and scholarly materials referenced in this dissertation are properly cited and acknowledged.

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AJIBADE PRAISE TITUS

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Date

CERTIFICATION

I certify that this project work titled “RETROFITTING OF A MANUAL WHEELCHAIR INTO AN ADAPTIVE VOICE CONTROLLED HYBRID MOBILITY SYSTEM FOR PAEDIATRIC MOTOR-IMPAIRED USERS” is an original research work carried out by AJIBADE PRAISE TITUS (2020/9415) in the Department of Mechatronics Engineering, College of Engineering, under my supervision. I have examined.

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DEDICATION

This project is dedicated to God Almighty, whose wisdom, guidance, and grace have been my strength throughout this journey.

I also dedicate this work to my parents who endlessly loved, made personal sacrifices and have been very instrumental in my development both academically and personally.

Special thanks to my sister whose support and encouragement have been a great source of motivation and inspiration.

ACKNOWLEDGEMENTS

I wish to thank very much the Almighty God, His guidance, wisdom, persistence and grace during this project. He is the backbone of my support at each of the phases in this process.

I would like to sincerely thank my supervisor, Engr. Daniel Akinwande for his contribution and constant guidance to my work.

I would also like to appreciate my Head of Department Dr. Ezugwu Chinedu and his leadership and constant support in the Department of Mechatronics Engineering.

My heartfelt appreciation goes to my colleagues, whose collaboration, insightful discussions, and shared experiences have made this journey both enriching and memorable.

I am deeply grateful to my uncle, Mr. Peter Adeyemo, for his unwavering support and encouragement. His belief in my potential has been a major source of motivation.

Lastly, I extend my profound appreciation to my family, whose love, sacrifices, and prayers have been my foundation. Their constant support has been the driving force behind my achievements.

To everyone who contributed in one way or another to the success of this project, I say a heartfelt thank you.

ABSTRACT

This report presents the design and implementation of a voice-controlled wheelchair with hybrid control, tailored specifically for paediatric users with motor impairments. The system integrates both voice recognition and manual control via a two-axis joystick to ensure safety, reliability, and consistent accessibility. Voice commands such as "move forward", "turn left", and "stop" are processed through an offline Android application developed using the Massachusetts Institute of Technology (MIT) App Inventor, which enables low-latency command execution without dependence on constant internet connectivity. The joystick serves as a backup in case of voice recognition failure or communication loss. The methodology involved constructing the control system on two microcontroller platforms: the Arduino Uno for hardware-level control and the Node Microcontroller Unit (NodeMCU) for wireless communication. Performance evaluation focused on command reliability, communication responsiveness via Wi-Fi, mechanical robustness, and power stability. Experimental results revealed a voice recognition accuracy of approximately 90% in quiet environments, and the joystick input remained responsive under all tested scenarios. Mechanical issues such as shaft-to-wheel slippage and inconsistent chain tension were corrected through in-field adjustments, resulting in improved drive performance during directional manoeuvres. Overall, the system demonstrated the feasibility of a cost-effective, dual-mode assistive mobility solution that can significantly improve autonomy and independence for children with physical disabilities.

Word Count: 207

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

The evolution of wheelchairs has been pivotal in enhancing mobility and independence for individuals with physical disabilities. The concept of wheeled seating devices can be found in historical records of ancient civilizations. People have developed wheelchair technology through centuries and scientists initially documented these prototypes in ancient China and Greece during 2013 according to Ability Tools. A 6th century Greek ceramic vase presents a picture of a wheeled child's bed demonstrating primitive efforts to enhance mobility functions (A Brief History of the Wheelchair, 2021). In 1595 King Philip II of Spain who suffered from severe gout used a chair with tiny wheels as one of the initial documented wheelchair devices for personal movement during that time. The wheelchair industry experienced strong development during the 17th century. The German inventor Stephan Farfler designed a self-propelled chair with a three-wheel framework which gave independent navigation capabilities for users in 1655. The invention provided a foundation for upcoming advancements in portable accessibility devices (Kay Nias, 2019).

During the 20th century scientists made major advancements in wheelchair technology while developing its performance abilities. The world received its first lightweight collapsible wheelchair during the 1930s when American engineers Harry Jennings and Herbert Everest introduced this historic wheelchair model (A Brief History of the Wheelchair, 2021). The new wheelchair design marked the beginning of today's user-friendly manual wheelchairs which focus on user independence. Electric-powered wheelchairs added an innovative option for

people with disabilities to enhance their mobility capabilities. New wheelchair designs included electrical systems with motors and battery capabilities which eliminated the need for intense manual effort and allowed users to explore various ground conditions more easily. User autonomy along with life quality improved after technology became integrated with mobility devices.

Yet in recent decades significant advances have been made in wheelchair technology, especially in connection with the creation of smart wheelchairs. The existence of smart wheelchairs has been able to offer enhanced mobility and independence to persons with physical disabilities (Mohammed Hayyan Alsibai et al., 2015). These sophisticated devices are equipped with sensors, control systems, and user interfaces designed to assist individuals with severe mobility impairments. Smart wheelchairs can detect obstacles, navigate predefined paths, and adapt to various environments, thereby augmenting user safety and independence (Abdullah et al., 2021). Smart wheelchairs were originally defined as mobile robots endowed with seats, Simpson (2005) states that the evolution of smart wheelchairs has been confined in the arena of changing commercially available power wheelchairs to make them more accessible and adaptable for the needs of an individual user.

Different navigation approaches in smart wheelchairs help users through accessible options for various capabilities and choices. The main assistive function of smart wheelchairs occurs during emergencies to make corrective actions yet maintains emphasis on user independence. The adaptive shared control system allows performance-driven distribution of system control since Mounir Bousbia Salah et al. (2005) introduced this method. Semi-autonomous systems help users execute brief navigation commands, but independent navigation emerges from fully

autonomous systems which connect AI to sensor-based control points for pre-defined destination routes.

The progress of smart wheelchair technology has not extended evenly toward products appropriate for paediatric patients. The current wheelchair technology available on the market targets adult users because it does not accommodate essential child-specific features regarding vehicle dimensions, operational complexity, and weight capacity. Adult-oriented portable support systems create an urgent necessity to develop affordable child-friendly accessible mobility solutions which meet the special needs of young users.

Recent campaigns direct their efforts toward retrofitting manual wheelchairs through essential feature modifications while keeping these upgrades inexpensive and practical. Thanks to current developments in voice control technology and safety systems various manual wheelchairs have become easily attainable solutions for this issue. The integration of technology into existing manual wheelchairs establishes a cost-effective method to build user-specific wheelchair systems based on current manual chair structures. The documented approach provides cost-effective ways to modify manual wheelchairs so that more user groups can access these products (Barriuso et al., 2018). Users can operate wheelchairs through voice recognition technology that decreases their requirement for physical strength and dexterity (Manoj et al., 2017).

The two main voice recognition systems for wheelchairs include Speaker-dependent systems that need user training but optimize performance for each individual and speaker-independent systems that recognize multiple voices without training. The accuracy of speaker-dependent systems surpasses speaker-independent systems, but speaker-independent systems deliver better access to users (Abed, 2015).

The development of young children depends heavily on their independence for movement. Self-mobility enables children to discover their surroundings through interaction with others and develop their minds in significant developmental stages. The installation of accessible adaptable mobile aids for children with severe mobility issues creates significant improvements for their developmental progression along with their life quality (Barnett & Belfield, 2006).

This study develops a budget-friendly solution by installing voice-operated systems and other required units onto current manual wheelchair models. The current project attempts to address gaps in accessibility by addressing them directly in a resource-efficient manner. The programme focuses on paediatric populations with severe mobility restrictions, to provide them with wheelchairs, which are, at the same time, affordable and accessible and based on the principles of user-centred design (Bray et al., 2020).

Retrofitting the manual wheelchairs on child-scale with voice-activated aid as a part of the project means that the project answers to two trends: it supplements the transportation capacity of the user, and it provides an individual adjustment of the device at the same time. Competent research and development work is the key to advance such technologies in such a way that they meet rising demands of the users, maintain a high level of safety, and achieve high reliability (Simpson, 2005). The history of assistive devices has always been aimed at the improvement of the movement possibilities of the disable people with the help of the wheelchair. Technological advancement has rendered simple wheeled chairs into advanced smart wheelchairs thus making it one of the major advances in life-changing products.

This exploration and the results achieved by retrofitting the standard manual wheelchairs with components of voice-control and other systems of battery and battery-management technology,

as well as the obstacle-detection systems, add to the future development and provide a real solution that allows even young children with severe mobility problems to move throughout their environment with better autonomy and confidence.

1.2 Statement of the Problem

Children diagnosed with tetraplegia and cerebral palsy along with spinal cord injuries and muscular dystrophy experience movement difficulties because they cannot use standard wheelchairs. The inability to use their upper body functions and motor skills prevents children from moving their manual wheelchairs, which cuts off their independence for mobility.

Children who have severe disabilities must depend completely on caregivers, parents, or family members to move their wheelchairs through the environment. Their over-reliance on others for assistance results in reduced autonomy while limiting social connections, educational and recreational participation which negatively impacts their emotional and psychological state.

Advanced smart wheelchairs featuring autonomous navigation, voice control and obstacle detection systems are available in other countries, yet they are inaccessible to most Nigerians due to their excessive cost. These wheelchairs are extremely expensive costing millions of naira, making them unaffordable particularly in a developing economy like Nigeria.

Retrofitting manual wheelchairs into smart voice-controlled wheelchairs for children could be seen as a solution to this problem. By the integration of advanced technologies such as voice control, obstacle detection, and autonomous navigation, these wheelchairs can empower users to regain some level of control over their mobility and reduce their reliance on others.

1.3 Aim and Objectives of the Study

1.3.1 Aim of the study

The aim of this study is to retrofit a manual wheelchair into a voice-controlled hybrid Mobility system for paediatric Motor-Impaired users.

1.3.2 Objectives of the study

The specific objectives of this project are to:

i. retrofit a manual wheelchair by integration of motorised movement for independent mobility, voice-controlled navigation for hands-free operation and an efficient power system with a rechargeable battery for extended operation.
ii. add an alternative control by the implementation of joystick control
iii. include an emergency stop mechanism for enhanced safety and accident prevention
iv. test and evaluate the performance of the wheelchair in real-life conditions, assessing accuracy, and other required parameters and ease of use.

1.4 Scope of the Study

The following paper undertakes retrofitting of a standard manual wheelchair to voice-controlled hybrid over solid traction to be utilized by children with severe mobility disability. Under such retrofitting, users have the privilege to control the necessary wheelchair (forward, backward, left, right, and stop) via the use of voice command though the joystick is activated as needed. Because such power source is installed as a rechargeable power source and is incorporated with the wheelchair, the shortages of time in performing demanding chores between recharges can be solved by periodical battery update. The system is an indoor wheelchair that has a little ability to move through even the smooth outdoor surfaces. The retrofitting services carry out to achieve

optimum performance of the individual model of the manual wheelchair is provided by special persons who are capable of choosing and fitting right components with the level of precision. The analyses of hybrid wheelchair are carried out at the hospital and rehabilitation facilities, which also undergo an evaluation of applicability, stability, as well as efficacy. The project will help the child who has limited mobility by providing a system that can be used in an independent manner by making it more affordable with the flexibility of the system to ease their accessibility.

The emergence of voice commands in the intelligent wheelchairs has had a significant impact on the field of assistive mobility, but there are various weaknesses that should be given scrutiny. These limitations affect functionality, usability and adaptability, therefore, pointing out the ways to recalculate in the future. The most important is the correctness of the works of speech recognition, particularly against background noise. This system relies on preset voice commands and as such, it is most likely that they would be misinterpreted or fail completely when speech is not clear or distorted. This shortcoming is particularly problematic, when it refers to children with speech problems or people that do not pronounce words regularly. Second, charging and battery life are some other factors. It is estimated that one and two hours on a charge will be the lifetime of the wheelchair, which is not enough to cover up people who will need to use the wheelchair throughout their days. Frequent replenishment may be problematic, as may be the availability of charging stations, especially in low-income countries. Third, mobility is influenced by the limits on the adaptability of terrain. The wheelchair has been optimized to outdoor smooth spaces and interior, which might be tricky to navigate rough terrain, steep slopes, or hard to navigate spaces. Lack of the functionality of climbing stairs also inhibits accessibility to specific settings. Last, there is an issue of cost and scalability. Even though the wheelchair is projected to be cheaper than the currently available smart wheelchairs, the higher specs like AI

navigation and use of LiDAR blocks to detect obstacles may significantly increase the production price and hinder mass use.

The new invention of a voice-enabled intelligent wheelchair is a significant contribution towards making life more mobile to children with serious physical impairments. In applying voice recognition, motorized movement, and security protocols, the system presents a hands-free mobility system and supposes an alternative to a standard joystick, hence independency, ease of operation, and adaptability. Even though the present prototype has a number of limitations, the project sets a foundation of future developments in assistive mobility technology. Improvements on accuracy in speech recognition, battery lifespan, and mass production at a low cost may further improve the usefulness and performance of the wheelchair. The iterative research, which will include a lot of field testing, together with constant feedback by the users, will be critical in the refinement of the system as well as making it fulfil the requirements placed upon it by its intended users. This paper therefore contributes in making the new science of smart assistive technology because it proves an economical and feasible approach to resolving the issue of mobility among children. With further development, this technology can help create more inclusive environments, ensuring that children with mobility impairments can navigate their world with greater independence.

1.5 Significance of the Study

This project aims to develop a cost-effective wheelchair assistive mobility system suitable for children with severe mobility limitations through manual wheelchair modifications that incorporate voice commands, joystick, and motor controls. The automated wheelchair system removes manual driving demands from users which results in expanded freedom and easier

operation for young wheelchair dependents who need support. Voice recognition systems within wheelchairs demonstrated effective mobility solutions for disabled individuals because they decrease caregiver support and allow independent transportation (Abed, 2015). This project supports worldwide initiatives which focus on developing assistive technology devices for specific user needs to create safer and more functional products.

Additionally, this study contributes to engineering and assistive technology research by exploring the feasibility of low-cost retrofitting solutions for existing mobility aids. The findings could serve as a foundation for future improvements, including enhanced battery efficiency, additional control options, and greater adaptability for different wheelchair models.

CHAPTER TWO

LITERATURE REVIEW

The literature review serves as the cornerstone of this study through its research exploration into smart wheelchairs combined with voice control systems and paediatric mobility solutions along with their existing technological advancements and known gaps. The foundation for building voice-controlled smart wheelchairs is defined in this section by showing why children with severe disabilities require improved mobility. This paper traces wheelchair development starting from basic mobility aids all the way to contemporary high-tech assistance systems. The research tracks smart wheelchair development while reviewing voice command integration and explains the difficulties children encounter during mobility. Safety considerations together with usability tests and power management evaluations of smart wheelchair systems appear within this research to verify a practical and efficient proposed design. A person's autonomous movement status shapes their participation in both education and social involvement and routine functions independently.

A growing number of studies indicate independent mobility enhances child development by producing positive effects on both mental and social aspects and autonomy from caregivers (Simpson, 2005). Severe physical disabilities stand in the way of using conventional wheelchairs for these patients since the equipment requires upper-body strength and is often too heavy and large (Alsibai et al., 2015). The excessive expense of powered wheelchairs together with their sophisticated nature maintains barriers to independent mobility for numerous families according to Barriuso et al. (2018).

The field of assistive mobility technology has progressed, yet it lacks accessible and suitable smart wheelchairs for children. Available industry models are created for adult use, so they include adaptations which young users find too intricate, too large, and too expensive to consider. Research studies demonstrate that using affordable smart technology to retrofit manual wheelchairs allows wider access to wheelchair systems through cost-effective approaches (Abed, 2015). The functionality of voice control systems has emerged as an acceptable substitute for joystick-controlled powered wheelchairs to provide hands-free operation for individuals who have limited mobility in their upper body (Mounir Bousbia Salah et al., 2005).

The research investigates past work concerning smart wheelchairs and voice control systems to evaluate their present designs through analytical assessment of their benefits and drawbacks. The evaluation of technological patterns with safety considerations and realistic application barriers will demonstrate established practices for smart wheelchairs specifically designed for children. The review implements a systematic methodology that includes research materials from peer-reviewed journals and conference papers and published case studies to support its comprehensive findings. The research demonstrates the necessity to develop a paediatric-friendly voice-controlled smart wheelchair because existing technological limitations require attention. A device which fulfils the requirements of being affordable yet safe and efficient should be developed because it would improve mobility for children with severe disabilities. The research findings collected in this review will guide prototype development for an economical smart wheelchair by using evidence-based user requirements and practical needs assessment.

2.1 Historical Evolution of Wheelchairs

Through the history of wheelchair development humans have continuously worked to give individuals with disabilities improved mobility and autonomy. Primitive design has evolved through multiple stages to produce the advanced modern wheelchair devices we see today.

2.1.1 Ancient origins

Wheeled mobility devices have their origins in ancient historical periods. Archaeological evidence demonstrates that the first documented wheeled furniture appeared through a Chinese stone slate inscription and a Greek vase frieze showing a child's bed during the period between 6th and 5th centuries BCE. The artifacts reveal wheeled device usage, but these devices were not developed for mobility-impaired individuals (Herman, 1969). During the period of 525 CE in China artists started depicting specialised wheeled chairs intended for human transportation which marked an important milestone in personal mobility assistance (Watson and Woods, 2015).

2.1.2 16th and 17th centuries: Early innovations

During the Renaissance era designers made substantial progress in wheelchair technology development. A Spanish inventor created a complex chair for King Philip II in 1595 because the monarch suffered from gout. The chair from 1595 incorporated both armrests and leg rests but needed help from others for movement because it lacked proper propulsion features (Wolfson, 2014). At age 22 Stephan Farffler constructed the first wheelchair as shown in Figure 2.1, that operated independently when he built it in 1655. The three-wheel chassis vehicle combined cranks with cogwheels to enable independent user movement. The device served as a handcycle instead of a modern wheelchair because it featured hand cranks attached to the front

wheel according to Watson & Woods (2015). Around the same period, Sir Thomas Fairfax who commanded in the English Civil War used a self-propelled wheelchair. The injuries he received in combat forced Fairfax to use a self-propelled wheelchair as he lived through his last years. The wheelchair of Sir Thomas Fairfax can be viewed at the National Civil War Centre in Newark-on-Trent (Stockley, 2024).

2.1.3 18th and 19th centuries: The bath chair and beyond

John Dawson created the "Bath chair" in Bath, England in 1783 marking the start of the late 18th century. The three-wheeled device had two large rear wheels and a small front wheel which an attendant operated through a long handle for steering (Mary, 2018). A picture of the Wickerwork Bath Chair is shown in Figure 2.2. The United States introduced wheelchairs under the name "rolling chairs" for the first time in 1887. This seating option initially served invalid tourists on the Boardwalk, but healthy tourists soon adopted these decorated chairs by hiring servants to push them for display of their luxurious lifestyle (Dick & Sonu, 2013).

2.1.4 Early 20th century: Folding wheelchairs

The invention of the first portable steel folding wheelchair as shown in Figure 2.3, happened in 1933 through the work of mechanical engineers Harry C. Jennings Sr. along with Herbert Everest who developed the design. The X-brace design from Jennings Sr. and Everest transformed wheelchair storage and portability. The design created by Harry C. Jennings Sr. and Herbert Everest became the base for contemporary manual wheelchairs and continues to impact the industry today (Watson & Woods, 2015).

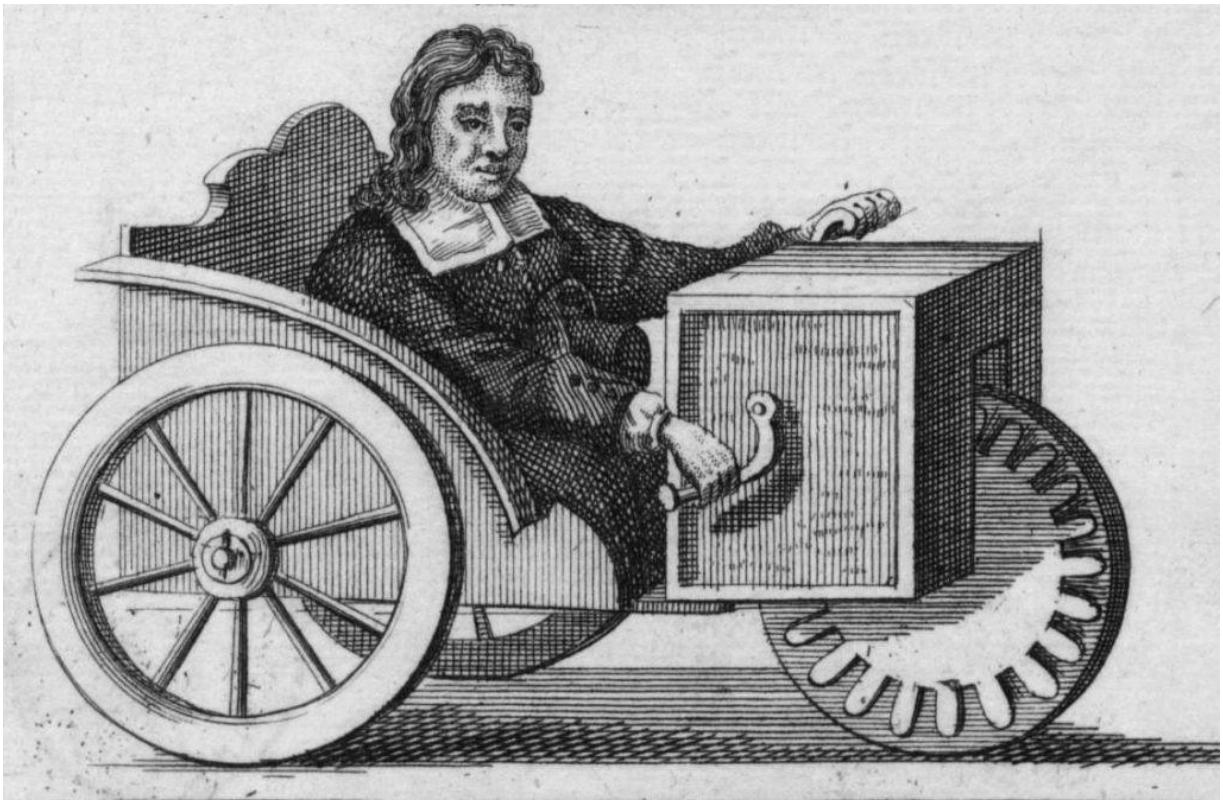


Figure 2.1: Engraving from c.1730 showing Fafler in his 50s sitting in his vehicle (Source: Science Museum Blog)



Figure 2.2: Wickerwork Bath Chair, English, early 20th century (Source: Science Museum)



Figure 2.3: Folding wheelchair, ‘Model 8’, English, c.1958 (Source: Science Museum Blog)

2.1.5 Mid-20th century: Powered wheelchair

Powered wheelchairs emerged in the middle of the twentieth century to meet the needs of people who could not propel themselves. Inventor George Klein from Canada first designed the power chair, which engineers and patients together created by sharing their requirements with medical professionals. Everest & Jennings became pioneers in mass production of powered wheelchairs, but their early models provided only minimal speed choices and basic steering capabilities (Erin, 2023).

2.1.6 Late 20th to early 21st century: Technological advancements

During the late 20th century specialized sports wheelchairs became available to enhance athletic mobility for people who use wheelchairs. The combination of lightweight materials with athletic designs resulted in equipment that allowed better participation in basketball and tennis and other sports as described by Erin (2023) and it is shown in Figure 2.4.

Smart wheelchairs combine advanced technologies including artificial intelligence (AI) and sensors with voice control systems which entered the market in the 21st century. Such innovative developments focus on improving user autonomy and safety to enable better environmental interaction, which shows an ongoing commitment to better mobility solutions (Erin 2023). The evolution of wheelchairs from ancient, wheeled furniture to modern smart devices illustrates a remarkable journey of innovation driven by the desire to improve mobility and quality of life for individuals with disabilities. Each advancement has built upon previous designs, incorporating new materials and technologies to meet the diverse needs of users.



Figure 2.4: ‘Shadow Racer,’ lightweight sports wheelchair, West Midlands, England (Source: Science Museum)

2.2 Technological Advancements in Smart Wheelchairs

Wheelchair technology has evolved throughout the last few decades from simple mobility aids into intelligent devices with sensor systems and AI assistance which enhance accessibility together with safety and independence. The implementation of artificial intelligence (AI) technology and sensor networks and automated navigation and advanced user interfaces with voice control and eye tracking and brain-computer interfaces (BCI) defines smart wheelchairs. New wheelchair technologies exist to assist people with major mobility limitations who cannot operate standard manual or powered wheelchairs. According to Simpson (2005), the integration of machine learning algorithms along with obstacle detection systems and energy-efficient power management systems has become the focus for enhancing smart wheelchair autonomy, adaptability, and usability. Researchers keep developing cost-effective scalable solutions for mobility assistance because population aging coincides with rising demand for these devices (Alsibai et al., 2015). This diverse levels of automation in smart wheelchairs, the various control mechanisms, sensor integration and AI-powered navigation, safety features, power efficiency, and the future directions of smart wheelchair technology are reviewed.

2.2.1 Levels of automation in smart wheelchairs

The automation level of smart wheelchairs controls the extent to which the system manages movement and navigation functions. Three levels of automation exist with specific benefits and constraints that users should consider:

Manual Wheelchairs with Assistive Features: The operation of traditional manual wheelchairs depends on both push movement done by users with their hands and aid from their caretakers for movement. Modern wheelchairs now include assistive systems including auto-adjustment features in addition to automatic brakes and power-assisted wheel controls to aid user mobility.

Partial automation systems in these devices enable smooth safe operation for users while maintaining their control over the device (Fehr et al., 2000).

Semi-Autonomous wheelchairs: The automation provided by semi-autonomous wheelchairs lets users control their devices alongside receiving aid for navigation and detecting obstacles while the system plays an active role in route planning. AI-assisted wheelchairs would push away from barriers alongside user-directed wheelchair controls. They also suggest alternative paths to navigate space without removing user control from the device. Some systems, such as the "shared control" model, combine user input with AI guidance to enhance safety without completely overriding user autonomy (Huo et al., 2008).

Fully Autonomous Wheelchairs: The automated wheelchair system operates without user supervision because it functions as an autonomous device. It usually consists of a combination of LIDAR, ultrasonic sensors and depth cameras with GPS and AI-based path planning technology that allows autonomous movement between locations (Wang et al., 2020). The autonomous wheelchair system perfectly suits people who have complete motor impairment including quadriplegia because it operates without any human intervention.

2.2.2 Control mechanisms in smart wheelchairs

The performance of smart wheelchairs depends on their control system which enables users to drive and steer the device through functions that adapt to their physical and cognitive abilities. The lack of suitable control methods for all users led to the creation of new interfaces for wheelchair operation. These systems are created to handle users who need assistance due to severe motor problems and speech difficulties and cognitive disabilities.

The joystick remains a common control interface in powered wheelchairs. It allows users to control direction and speed through hand movement. Users with restricted hand strength benefit from force-sensitive joysticks according to Cooper et al. (2000) while Pajkanovic & Dokic (2013) show how head-controlled joysticks let users navigate through motion using head tilts instead of hand movements.

Pacnik et al. (2005) explains that voice-controlled systems operate through speech recognition technology to translate verbal instructions into wheelchair movement. The system provides exceptional benefits to wheelchair users who have quadriplegia or severe limb impairments. Users experience two main difficulties when using voice recognition systems because speech clarity issues affect children and people with neurological disorders, background noise creates interpretation problems along with delayed responses that depend on system processing speed. The Eye-Tracking Control system enables wheelchair control through display screen or predefined area observations using eye-tracking technology (Mohamad et al., 2016).

The control systems prove beneficial for patients who have severe motor disabilities including those with ALS. The system employs Infrared-based eye tracking to detect accurate gaze direction, and it has on-screen selection menus with directional icons for users to activate movement through eye observation. Also, the system features adjustable calibration that customizes to fit unique eye movements.

Users can operate the wheelchair through thought patterns which Brain-Computer Interface (BCI) Control translates into movement commands. Wheelchairs powered by BCI bring significant advantages to people with a locked-in syndrome or severe spinal cord damage or

possible neurodegenerative disorders (Halima & Hessa, 2018). These systems are operated through usage of Electroencephalography (EEG) sensors placed on the scalp that transfer neural signals to machine-learning algorithms that gets these signals translated to proxies of movement. Users get special training to sharpen control accuracy. The Sip-and-Puff Control system is created to accommodate persons who have limited or no control over their limbs. A tube-like straw contains sensors of pressure that reveal the signs of inhalation and exhalation hence producing the commands (Harkishan et al., 2018).

Depending on particular patterns of a sip-and-puff, users drive their devices to make some movements like forward, backward, left, and right, at a varying speed regulation by the intensity of breathing. Puffing sequence is the method through which an emergency stop motion is realised. The developed systems to control wheelchairs are based on the use of motion sensors and depth cameras to operate by a gesture; Sadi et al. (2022) detail one system that senses hand waves, as well as finger gestures or facial expressions.

The receiving of data is carried out through Kinect-based or Leap Motion controller, and the performance of commands is implemented with the help of hand gestures and facial expressions such as raising eyebrows to move forward and blinking twice to stop. The lack of manual controls makes this technology particularly beneficial to persons who are affected by different degrees of motor disability.

The technological improvements in the control systems of the smart wheel chairs have achieved appreciable elevation because of the incorporation of the various input modes overcoming a wide range of impairments. The adaptability features have been incorporated in the modern systems, which promote an increased operational level of user autonomy and freedom of

mobility. Its future directions include the need towards the combination of different control configurations, thus aiding people with or without cognitive and physical impediments to use wheelchairs with greater reliance, safety, and success.

2.2.3 Safety and collision avoidance in smart wheelchairs

The emergence of such smart wheelchairs has significantly increased the safety of the user as well as their ability of control over the movement equipment. The latest models combine complex sensor systems with artificial intelligence and ease of design controls to support safe, reliable and predictable collision-avoidance capability. Such systems must put safety considerations at the fore front in the conception and deployment of such systems. Since the nature of navigable terrains is highly complex, powerful safety systems are all necessary to safeguard people against unforeseen hazards. With the current technology, it has been easier to develop user-friendly systems that can identify obstacles in the environment and allow stability in maximizing personal mobility. The safety of navigation in smart wheelchairs depends on the complex of sensors, which are preconditioned to detect any obstacles as well as seamless movement. The entire sensing of the environment enables the activity to be conducted in various environments. There is a set of vital sensors which run as follows:

(i) LiDAR, uses laser pulses to produce extensive 3D geographic representations that are generated by the Light Detection and Ranging technology (LiDAR);

(ii) Radar sensors using radio-frequency identify the objects in the proximity with an angular resolution of 0.5 degrees;

(iii) Proximity sensors, which react to radar, recognize objects up to 15 centimetres from the wheelchair and have a spatial resolution of 0.2 centimetres;

(iv) Proximity sensors: the ultrasonic proximity sensor has a spatial resolution of 1.3 millimetres and can detect objects up to 40 centimetres.

In combination, these sensor modalities, according to their capacity, provide excellent environmental context awareness which is used to navigate in diverse environments with ease and safety. This technology delivers exact obstacle detection capabilities that work exceptionally well in challenging setting (Luiz et al., 2014). Ultrasonic Sensors transmit sounds waves which detect surrounding objects while simultaneously calculating their distance. The technology delivers dependable information which protects wheelchair users from hitting objects particularly in areas where precision is essential for avoidance (Luiz et al., 2014).

Infrared (IR) Sensors: Detect heat signatures and proximity of objects. Obstacles become more recognizable because this technology operates effectively during low light as well as dark situations. Real-time deep imaging emerges from depth cameras which amalgamate RGB (colour) and depth information. The accuracy of object recognition and navigation improves when a wheelchair obtains better environmental details about its surroundings.

Wheelchairs that implement powerful obstacle detection system work by uniting multiple sensors which enables safe movement and collision prevention. Multi-sensor integration enables the system to develop an important level of environment detection for its adaptation between indoor and outdoor spaces. A combination of multiple sensors remains vital for delivering dependable and safe navigation solutions particularly when serving children and impairment users of the wheelchair. Collision avoidance systems are also built into smart wheelchairs work to boost user protection while stopping incidents. Such wheelchair systems combine user-

controlled mechanisms with predictive analytics capabilities together with emergency response strategies to provide trustworthy navigation throughout various locations.

Emergency braking stands as a vital feature because it activates automatic stopping of the wheelchair upon detecting potential collision situations. This protective system functions as a backup feature that enhances wheelchair safety during dangerous situations. When the wheelchair moves toward an obstacle too fast or when the user does not respond quickly enough the system will activate to stop an accident (Martorello & Swanson, 2006).

Shared control systems act as essential components for preventing collisions while operating a wheelchair. User input and automatic corrections exist in harmony to let the wheelchair assist navigation while preserving user steering control. The system alters trajectory directions gradually when the user tries to steer toward obstacles so that collisions are avoided but user control remains intact. The system combination provides safety measures alongside user-controlled operation. The collaborative system functions to adjust operations while adapting to changing surroundings for safe and dependable performance. The system decreases accident risks and builds user confidence which allows people with mobility impairments to move independently in their environment.

Fall prevention and edge detection form two central pillars behind the safety of smart wheelchair systems. Such features are the ones that specifically play into minimizing the increased risks that users are exposed to in cases where the latter are confronted with stairs, curbs and other uneven surfaces. Knowing how to combine advanced sensors and stability systems, smart wheelchairs could maintain the balance and prevent accidents during tricky moves (Raed et al., 2020). Detection of steps and curbs the use of sensors, including LiDAR sensors, ultrasonic

transducers, and depth cameras, to identify elevation step-ups is referred to as stair and curb detection. In the event of a fall being identified by this type of feature, the system will trigger a haptic, light, or audible warning to notify the user. The wheel chair might automatically adjust its path to prevent the danger and protect its rider in some situations. In addition to these systems, there is a provision of tilt stability sensors that monitors persistently the orientation and angle of the wheelchair. In a case where a risk of lateral tipping is detected, the system reacts pro-active and corrects the situation by altering speed, braking and/or weight transfer in order to stay balanced. Such functionality is especially irreplaceable since outdoor settings often do not allow even and stable slopes. In sum, all these technologies make people safer and more confident when they may avoid dangerous situations and are free to move without any restrictions, so people can move through complex environments without any fear.

The security and efficacy of smart wheelchairs depend on the ability to investigate the health of the user and react to arising conditions. Devices of the kind utilize the all-inclusive platform in favour of general health by serving the purpose of mobility and possible health conditions. Facilities to make the user more secure that would increase safety and comfort include biometric tracking, emergency stop and carer alert systems. Vital signs including blood pressure, oxygen saturation, and heart rate are always automatically monitored using biometrics. Sensors that detect these variables embedded in them will notify caregivers and medical personnel in real-time any occurrences in parameters, e.g., an elevated heart rate or a test depletion of oxygen saturation. In line with health surveillance, emergency stop functions allow users to put the chair to a stop instantly, thus helping them sense a spirit of self-determination and feel much safer when it comes to paediatrics and individuals with limited mobility. Lastly, caregiver alert systems provide on-demand notification channel, most often through SMS, mobile apps, or

telecommunication solutions, in case of abnormal situations, i.e. falls, sudden break, or medical emergency, and thus, provide constant monitoring.

With the combination of the biometric care, emergency stops, and alerts on the side of caretakers, smart wheelchairs would provide a significant safety framework to the end-users. All these systems increase personal mobility at the same time challenging health and emergency aspects, making smart wheelchairs a reliable and safe alternative to persons with mobility disabilities. The mechanisms have a manual override capability and a failsafe mechanism that protects the users of these machines and makes them have control in case of anomalies.

These characteristics incorporate a new level of protection making patients or their caregivers able to intervene in case they see a need and that ensure that the wheelchair functions properly in any situation.

The manual override option allows the caregivers or any other attendant to take charge during times of emergency or system breakdowns. When the remote drive fails to navigate automatically or it seems to be hit with unseen obstacles, people can turn the wheelchair to manual mode. Finally, such a feature will make sure that users will never be left in no man no land or be unsafe due to technical issues.

In case of a manual override, smart wheelchairs have additional protocols that deal with system failures, making these failures as graceful as possible. Such procedures prevent the wheelchair from becoming uncontrollable in the event of software or hardware failure. For example, should some sensor fail or if the system's telemetry becomes disconnected, the chair can be programmed to decelerate, come to a halt, or disconnect from the main power supply. Such

measures reduce danger and allow people to use the wheelchair with utmost confidence, knowing it will act safely in unforeseen situations and conditions. Smart wheelchairs embrace manual override and fail-safe features for additional control and security. These mechanisms guarantee that users remain in control and safe, even when automated systems malfunction, which makes smart wheelchairs reliable for people with mobility challenges (Simpson, 2005).

Safety regulations and standards serve as a key pillar for smart wheelchair safety in development. Following these guidelines makes certain that the wheelchairs are used, constructed, and maintained safely, ethically, and legally, while also ensuring that they are durable and usable. Among the most important standards is the ISO Wheelchair Safety Standards which provides classification concerning wheelchair safety, performance, and durability on an international level. This includes considerations of stability, braking systems, and load capacity with an aim to make sure that wheelchairs are safe and reliable for regular use. It is equally important for wheelchair manufacturers to comply with ISO Standards as they set the benchmark of safety and quality for their products (ISO 7176-1, 2014).

Another important regulation is ADA Compliance, which ensures that the smart wheelchairs adhere to the safety and accessibility provisions of the Americans with Disabilities Act. This stipulation includes adjustable seats, easy to manipulate controls, and integration into the public infrastructure. ADA Compliance is extremely critical within The United States since it promotes the use of well-designed and effective mobility assist devices for every user irrespective of their ability (Jennifer et al., 2000). Through compliance with established safety regulations and standards, manufacturers can ensure their smart wheelchairs fulfil mandatory safety and accessibility requirements. Such compliance not only assures users of confidence but also that

the wheelchairs can be used in diverse environments and by diverse users, thereby promoting inclusivity and independence.

2.2.4 Battery efficiency and power management in smart wheelchairs

The functionality and reliability of a smart wheelchair is reliant on the power supply and energy management system. In contrast with conventional manual wheelchairs, smart wheelchairs depend on electric motors, sensors, microcontrollers, and assistive technologies that all need a continuous power supply. An efficiently optimized battery system guarantees the wheelchair's effective functionality for long durations while upholding safety, durability, and economic viability (Shivakumar et al., 2024). Development in battery chemistry, charging mechanisms, and energy recovery technologies has witnessed appreciable enhancement in the performance of electric-powered mobility aids. The most widely used battery types for motorized wheelchairs are Lead-Acid, Lithium-Ion, and Solid-State batteries, each with some unique benefits and drawbacks (John, 2025). Sealed Lead-Acid (SLA) batteries are the least expensive option; however, they are very heavy, low in energy density, and have long charging durations. Although the batteries are durable, their short runtime and lengthy recharging durations render them less appropriate for contemporary intelligent wheelchairs.

On the other hand, Lithium-Ion (Li-Ion) batteries provide the advantage of higher energy density, lighter weight, and higher charging efficiency and are therefore the battery of choice for contemporary assistive mobility aids. Li-Ion batteries usually last two to three times longer than lead-acid batteries and maintain a stable power output over their entire life. There is no way that we can even talk about the future direction of the assistive devices without understanding the nature of the battery technologies. The solid-state batteries, specifically, are a sought-after

change in this field of activity since they address the issue of thermal runaway and overcharging susceptibilities that is inherent in lithium-ion batteries. Through this they provide increased energy storage density, faster recharge rates, and increased safety margins as compared to their lithium-ion equivalents. These advances are able to transform the wheelchair market increasing the range, reducing the weight, and strengthening the stamina of smart wheelchairs.

The capability of undertaking these roles is achieved by enhancing the two areas of concerns; battery capacity and energy efficiency. Capacity is often given in ampere-hours (Ah) and means the time during which a battery can deliver power at a fixed constant current. Since most mobility devices use nominal voltages of 24 V, their battery pack can have a capacity range of 20 to 75 Ah. Greater capacity ratings are directly proportional to longer operation times without being recharged, so a 50-Ah battery, to cite an example, could be used to provide 50 A in one hour or 25 A in two. Such an ability is critical in maintaining uninterrupted indoor and outdoor functionality, especially to those users in whose daily usage requires the steady mobility experience.

A smart wheelchair's energy consumption depends on several factors, ranging from motor efficiency to terrain type, user weight, and usage patterns. On flat indoor surfaces, a wheelchair may consume up to 200 to 300 watts per hour, but rough outdoor surfaces or inclines can increase consumption to over 500 watts per hour. Additional loads and frequent starts, stops, or changes in speed also result in higher energy consumption. To alleviate these problems, smart wheelchairs often feature advanced technologies to optimize energy efficiency (Cooper et al., 1995).

Good motor control algorithms regulate power delivery to match current needs, thereby limiting power wastage and prolonging battery life. Adaptive power distribution systems adjust power delivery to various components, like motors and sensors, depending on current requirements. Regenerative braking is also found on some wheelchairs and is a feature that captures and recovers energy during braking, which is subsequently utilized to run the wheelchair. All these functions serve to optimize battery life and enable the optimal running of the wheelchair in a range of settings (Takahashi & Seki, 2009).

By prioritizing battery capacity optimization and energy efficiency maximization, manufacturers can develop smart wheelchairs with longer operating durations, less frequent recharging requirements, and improved overall performance. Smart wheelchairs, which stay longer in the field with their guaranteed safety and usability, are a reliable type of independent-mobility apparatus used by persons with different functional requirements. Charging systems hold one of the most vital engineering considerations of such devices, making a direct impact on both feasibility and daily usefulness. The conventional technologies usually utilize plug-ins chargers which send a constant current to recharge the batteries in about 6-8 hours. Although such set-ups are extremely dependable, they create considerable downtime, which virtually rules them out with populations that use the wheelchair as the sole means of daily transport. To curtail this limitation, the manufacturers have come up with fast-charging techniques, especially in the case of lithium-ion batteries.

Depending on the type of innovations made, some batteries can charge to an 80 % state-of-charge within one to two hours, much less than the current number of hours it takes to charge a battery, and, eventually, enhance the usability of the gadget in question. This feature is

particularly important in the person engaging in a very active lifestyle or living so in the wheelchair without a break. In addition to rapid charging, developers and scientists are exploring other design methods, namely, wireless chargers. They rely on inductive charging pads so that the user drags their wheelchair over a stationary surface to begin the charging process. This setup is especially helpful to individuals with poor hand movements or those who have weak hands since plug ins do not need any manual work. Wireless charging not only allows for convenience but also cuts down on wear and tear of the charging ports, increasing the lifespan of the wheelchair components. With the integration of fast charging and wireless charging technologies, it is possible to make smart wheelchairs more convenient and versatile for users. Such technologies see to it that users spend less time waiting for their wheelchairs to be charged and more time experiencing mobility and independence. As these technologies continue to advance, they are certainly going to be a part of future smart wheelchair innovations, further improving the user experience.

Another significant step in power management in smart wheelchairs is the application of regenerative braking, a mechanism that aims to recover and transform kinetic energy into electrical energy when the wheelchair decelerates or comes to a halt. In contrast to conventional braking mechanisms that lose energy in the form of heat, regenerative braking recovers some of this energy and deposits it in the battery. This process optimizes energy efficiency while at the same time extending the range of operation of the wheelchair, thus making it a vital feature for those who rely on their wheelchairs for extended periods (Takahashi & Seki, 2009).

Regenerative braking It can be especially beneficial during stop-and-go or rolling-hill traffic, with conventional application involving the loss of a large amount of energy. In a case of braking

down a slope, say, the kinetic energy produced in the braking process will be converted to electrical energy which is then used to recharge the battery. As empirical work suggests, it is possible to increase the lifespan of batteries of 10 20 % with the use of regenerative braking depending on the tendencies of use and on the terrain (Prasanth et al. 2023). Another example is that the technology will form a perfect solution to people who work in extreme environments and need a battery with long life. Regenerative braking and energy harvesting common in smart wheelchairs possess higher energy efficiency and sustainability (Prasanth et al. 2023). Such systems not only result in less frequent recharging but also contribute to the overall improvements in usability because it provides more reliable and eco-friendly mobility solution. With further technological advancement, it can be expected that these functions may come as cardinal elements of the upcoming smart wheelchair design, enhance its performance and enhance satisfaction among the end-users.

Clean energy technologies installed on smart wheelchairs (i.e., solar panels) are a new aspect of increasing battery capacity and energy efficiency. Though solar panels will not be able to substitute the main battery on the whole, they will provide them with a green energy-saving system that will allow enhancing the working range of the wheelchair by collecting more energy when the wheelchair is in the fields. The wheelchair frame or canopy can be fitted with flexible photovoltaic (PV) panels to enable it to trickle charge the battery, so reducing the dependence on conventional plug-in charging. Such an arrangement is very beneficial to the users with long outdoor days, as they can charge the battery on the move (Pita 2020).

However, present designs of solar wheelchair are trapped by a number of limitations. The efficiency of solar panel is subject to solar irradiance, the area of the panel and its internal

conversion rate. At the same time concomitant regular mobility-aiding wheelchair set ups have practical restrictions over solar-energy harvesting. As a result, solar powered wheelchairs can be more applicable in the provision of ancillary power aid, as compared to making the vehicle fully autonomous. The current practice is to use photovoltaic systems to charge the auxiliary elements; i.e. sensors, lighting, and communication equipment, and have the main battery power propulsion missions (Pita, 2020).

The limitations of this application would be met with future improvements in high power creating solar cells and future battery/energy storage technology that will increase the viability of solar powered wheelchairs. Energy yield per system could be increased significantly by enhancements in the efficiency of solar-panels and by producing lightweight and flexible panels. What is more, in battery-related terms, the improved capacity and higher charging speeds, combined with solar energy sources, may result in more functional and sustainable energy solutions. Use of solar technology in the manufacturing of smart wheelchairs will empower the manufacturers to offer consumers cleaner and energy efficient mobility products. Despite the constraints presented by the existing technology, the possibility of developing solar-powered wheelchairs further brings hope that they will further take root in the larger mobility context, which, in turn, will lead to the increased independence of people and fewer carbon footprints.

The need to continuously monitor and control the wheelchair batteries requires a Battery Management System (BMS) and hence, guarantees its safe and efficient operations. The BMS guards against overcharging, over-discharging, overheating, by imposing tight control over charging and discharging cycles and the short circuit. The newer technologies utilize the concept of the predictive analysis with the help of AI to check the condition of the battery and predict the failure, making preventative maintenance possible (Tseng et al., 2018).

Some of the main functionalities of the BMS are: Voltage Regulation, a system that offers the stable supply of power thereby eliminating the risk through power fluctuations that causes damage to the electrical equipment; Thermal Protection is a system that monitors and regulates temperatures to prevent chances of fire and overheating;

Short-circuit and overcurrent protection devices that automatically switch off the system in case of power surges or electrical faults; and State of Charge (SoC) estimation; this is a system that provides accurate measurements of the levels of battery to prevent an abrupt shutdown of the system.

Future trends in wheelchair power systems are a sign of efficiency, sustainability, and robustness. There is particularly study on Hydrogen Fuel Cells, where electricity is created by the wheelchairs with water as the only by-product. This method has increased range and quicker getaway than customary battery plans.

Graphene Batteries: Graphene batteries are touted to provide faster charging, higher energy density, and significantly longer lifespan compared to lithium-ion variants.AI-Powered Energy Optimization: Energy management systems based on artificial intelligence will dynamically optimize power distribution based on real-time consumption patterns, providing maximum efficiency.

Efficient battery management and power optimization are crucial for the performance and reliability of smart wheelchairs. With advances in lithium-ion and solid-state batteries, fast charging systems, regenerative braking, and AI-powered energy monitoring, smart wheelchairs are becoming more efficient, reliable, and sustainable. As new energy storage technologies

emerge, wheelchair users can expect longer battery life, reduced charging times, and improved safety in future mobility solutions.

2.2.5 Future trends in smart wheelchair development

Smart wheelchair development experiences rapid progress because of new advancements in artificial intelligence (AI) and Internet of Things (IoT) and robotics as well as innovative material applications. Smart wheelchair technology aims to provide better mobility capabilities and self-governing ability and better life quality to people with disabilities.

Smart wheelchair technologies with integrated AI functions learn adaptive control capabilities because they perform predictive analytics to personalize reactions for specific user actions. AI technology enhances user security by adapting to preferred destinations and utilizes adaptation algorithms to better match user behaviours for optimized efficiency. Voice recognition technologies have evolved to bring users better control of their wheelchairs through command systems that imitate natural language communication.

Users now possess the ability to operate wheelchairs directly by using brain signals instead of conventional physical controls. The electroencephalography (EEG) technique detects brain signals that indicate thoughts or intentions which get converted into instructions that operate the wheelchair. The innovation delivers special advantages to people with severe motor impairments because it enables them to gain unprecedented control of their mobility systems (Shiva et al., 2024). The combination of smart wheelchairs with IoT and smart home technologies produces connected living spaces which provide better support for users. The integration of IoT technology in wheelchairs allows users to manage their home automation systems through their mobility device for controlling lighting as well as temperature and

appliances. Users benefit from this integration because they gain independence and convenience to operate their environment without requiring help from others (Elvis, 2024).

5G technology enables quick data transmission and fast communication processes that link smart wheelchairs to cloud-based services. The wheelchair maintains optimal performance through real-time navigation updates and remote diagnostics and over-the-air software updates because of its connectivity to cloud-based services. The advanced features of telemedicine consultations and healthcare monitoring system integration are possible because of enhanced connectivity.

Through materials science innovations scientists create wheelchair frames which become both stronger and lighter. Carbon Fiber alongside graphene enables the production of advanced composite materials which create lighter wheelchairs that maintain their structural integrity for improved manoeuvrability. Morphing wheels represent a major wheelchair advancement because they can change shape to get over barriers such as curbs and stairs thus enabling users to travel across more terrain types (Hyun et al., 2024). The integration of autonomous navigation systems that utilize AI-based processing in combination with sophisticated sensors permits smart wheelchairs to complete difficult path planning operations without heavy user interaction. This technology monitors obstacles in addition to creating efficient paths and adapting its operation when surroundings change for better protection in unsafe situations. Wheelchairs that integrate with smart city infrastructure establish communication channels with urban elements such as traffic lights and public transportation, so users experience easier and more efficient travel in urban environments (Tripathy et al., 2020).

Smart wheelchair power systems undergo design changes because of the growing demand for sustainable energy solutions. The development of alternative energy solutions involving hydrogen fuel cells and graphene-based batteries seeks to produce enduring powerful systems for smart wheelchair applications. These technologies provide extended operational distance while lowering pollution effects from traditional battery creation and waste management (Choudhury & Gogoi, 2024).

Advancements in technology have led to increased efforts for developing smart wheelchairs that are both accessible and affordable for users. The rising demand together with economies of scale in mass production leads to declining costs. Policy initiatives together with funding programs are actively supporting people to buy advanced mobility devices so more populations can benefit from modern wheelchair technology. Smart wheelchair technology moves forward through multidisciplinary research of artificial intelligence together with materials science and power solutions and connectivity networks. New innovations work together to enhance wheelchair functionality as well as comfort and accessibility which creates improved quality of life for users. Research progress and development will drive smart wheelchairs into the main assistive technology framework which will deliver unmatched independence and mobility to people with disabilities.

2.3 Voice Controlled Wheelchairs

Voice-controlled wheelchairs represent a substantial progress in assistive technology because they allow users with mobility disabilities to direct their wheelchairs through voice commands. The innovation uses speech recognition technology to convert spoken words into wheelchair movement which gives users independent control and eliminates the need for manual controls.

The systems offer exceptional benefits to people with restricted upper limb capabilities by allowing them to navigate through voice commands (Clark & Roemer, 1977).

2.3.1 Types of voice control systems

Wheelchair voice control systems operate under two main categories as speaker-dependent and speaker-independent systems. A speaker-dependent system needs voice training from a specific user before it can accurately recognize commands. These systems succeed in reducing errors caused by accents alongside speech impairments and pronunciation differences because they learn the user's distinctive speech patterns during the setup process. The need for training creates an obstacle to immediate system usage (Kim et al., 2023). Other systems function as speaker-independent systems because they can detect commands from any user who has not trained the system before them. The effectiveness of the existing LLM-based models is reduced when considering the circumstance where various accents or phonetic disorders are present, and therefore, the models are not very useful in cases of people whose speech does not fit into the standard frames (Kim et al., 2023).

2.3.2 Speech recognition challenges

There is a large element of technical impediment to voice control systems mounted on the wheelchair. The main three variables that catalyse the effectiveness of the system are the background noise, user speech intelligibility, and the ambient acoustic conditions (Masina et al., 2020). High ambient noise in the environment causes impaired recognition and incorrectly perceived commands and possible safety risks (Masina et al., 2020). Individuals have to learn

to recognize different recognition patterns and people with speech disabilities face even more barriers on the way to providing effective command recognition (Masina et al., 2020).

2.3.3 AI and machine learning in voice control

The combination of artificial intelligence and machine learning technologies has improved the features of voice-controlled wheelchairs. Through user interaction the machine learning algorithms help the system gain knowledge that leads to better command recognition while learning distinct speech patterns of individual users (Hariharan et al., 2025). Also, through AI technology the system gains the ability to understand context which results in better natural interactions with fewer chances of misinterpretation (Hariharan et al., 2025). Machine learning uses patterns to forecast user intentions which leads to predictive assistance that provides tailored navigation experiences (Hariharan et al., 2025).

2.3.4 Studies on voice controlled assistive mobility

Scientific investigations have shown that wheelchair control through voice commands presents several possibilities. The initial research phase integrated voice recognition systems into wheelchairs which required solutions for power efficiency and system dimensions to make them useful for wheelchair users. (Clark & Roemer 1977). Recent research combines voice control with touch or gesture controls to create safer and more reliable systems which support users in various situations (Priyanayana & Jayasekara, 2018). AI integration with computer vision technology allows facial expression recognition systems to function as commands thus creating new access methods for users who struggle with speech (Yacine et al., 2023)

2.3.5 Comparison with other control methods

Voice control produces specific benefits together with specific constraints when compared to alternative wheelchair control methods. The use of joysticks through Manual Controls demands precise motor skills from users yet such abilities might be hard to achieve by people with upper limb disabilities. The system's voice control feature enables users without need for manual input to operate it (Kim et al., 2023). Gesture controls provide intuitive operation yet experience problems with precision and need unblocked sensor views which makes them impractical in some circumstances (Rahimunnisa et al., 2020). The implementation of Brain-Computer Interfaces (BCIs) for neural signal control requires invasive procedures and complex calibration which limits their practical application for wide-scale adoption. Voice control functions as a convenient and contact-free solution for users (Kim et al., 2023).

Voice-controlled wheelchairs have completely changed assistive mobility through their user-friendly navigation systems designed for people dealing with mobility issues. Despite current hurdles with speech recognition precision and environmental interference AI along with machine learning technologies continue to improve wheelchair systems' stability and usability capabilities. The solution of these challenges will allow voice-controlled wheelchairs to enhance user autonomy which leads to better independence and improved life quality.

2.4 Empirical Study on Voice Control and Hybrid Wheelchairs

Research on the area of developing intelligent mobility aids to people with disabilities, most importantly to paediatric users, has now gained relatively high empirical research interests during the past few years. These researches dwell on the enhancement of control systems, safety, cheapness, and flexibility. This subsection will review some of the empirical literature that will

be taken into consideration in the context of the current project in adaptive retrofitting of a manual wheelchair to a hybrid voice-controlled wheelchair.

Hariharan et al. (2025) have practically implemented a voice-controlled wheelchair suitable to the needs of both the visually impaired and physically disabled users using a speech recognition module coupled with a micro controller Arduino. Their model responded to commands of simple movements (“forward”, “backward” and so on) through voices, which were translated into motor actions. The experiments indicated that the low-noise environments achieved a recognition accuracy of 87 percent yet it was poor when there was an outdoor or noisy situation. It was noted in the study that fall back control modes such as joystick, manual override was important to use when voice input fails, and this falls in line with the hybrid approach taken in the project.

Ramaraj et al. (2022) introduced a modular real-time shared-controlled framework that uses stereo camera arrangement and Robot Operating System (ROS) in smart wheelchair. They decreased the number of encoders by relying on visual odometry, so mapping, obstacle detection and planning of trajectories could take place in real-time. The system tested in the simulation and actual conditions was very efficient in manoeuvring in tight areas. Being not voice-based, they nevertheless showed how modular systems can make such products more reliable and allow more user autonomy, which is very important when it is applied to the situation with paediatric users and their spatial and developmental needs when adapting wheelchairs.

In their study, Joseph, Aswin, and Prasad (2019) built a wheelchair that responds to voice and gestures of the user augmented with ultrasonic sensors to detect obstacles. The study designed

and deployed on an Arduino Uno and motor driver modules was able to identify response to user inputs and command execution performance variables in the various environments. They have reported an 8592 percent command accuracy in recognition of voice and that, when fallback gestures were also incorporated, the satisfaction rate was improved. Their income experiment confirms to us the use of the dual-mode control system that is in use in this work.

The paper by Priyanayana and Jayasekara (2018) investigated a voice and joystick mixed multimodal control system that would provide increased safety. The voice module was an offline system with a memory key application on the smart phone, whereas joystick continued to be the manual override at all times. A test against ten participants with different levels of motor functioning proved that hybrid control minimized fear and boosted the sense of control. The results of their research provided substantial evidence of the idea that both voice and manual input must be maintained active the approach that is used in this system.

Ruiz-Serrano et al. (2014) integrated information about the direction of the smart wheelchair in multimodal navigation interface that uses tactile, visual and proximity sensors. The field test proved that the simplified, redundant controls enhanced stability in navigation and user satisfaction. Physical buttons and joysticks were significant in case the automatic mode did not work by users with low muscle tone or poor cognition. The reason is that their system specifically targets the children and old users whereas the existing design was paediatric with regards to the design.

Lastly, in the analysis or discussion by Masuzawa and Minami (2009) dealing with safety, they used an emergency-shutdown system on the joystick-operated electric wheelchair. When put

through some use cases, the system also guaranteed immediate power cut-off during the emergency, which demonstrated the necessity of such features in any type of assistive mobility system, including the voice-controlled ones.

2.5 Paediatric Mobility Challenges and Solutions

The ability for children to move independently remains essential for their development because it directly shapes their mental and social and physical growth. Children who face mobility issues need specific solutions which allow them to fully participate in daily routines and social communication.

2.5.1 Common mobility disorders in children

Multiple medical conditions lead to child mobility impairments among which cerebral palsy stands alongside muscular dystrophy and spina bifida and spinal cord injuries. The medical conditions produce weak muscles and spasticity and paralysis which prevents children from moving freely. Cerebral palsy stands as one of the most common physical disability causes in children since it produces disorders which affect movement control and muscle tone. The genetic disorders of muscular dystrophy cause ongoing muscle breakdown that generates added obstacles to movement (Kruer, 2015).

2.5.2 Limitations of Traditional Paediatric Wheelchairs

Children who use traditional wheelchairs encounter multiple obstacles which limit their freedom in daily activities and affect their involvement in different pursuits. Children with restricted muscle function struggle with using manual wheelchairs because these devices need strong upper body control and precise movements. Powered wheelchairs deliver increased mobility,

but they have high price points and weight limitations that make them unsuitable for customized child-use. These wheelchairs have dimensions which block accessibility in some settings and thus restrict children from moving freely through spaces (Diane, 2003).

2.5.3 Benefits of smart wheelchairs for children

Advanced technology integration in wheelchairs enabled the production of smart wheelchairs that bring multiple advantages to paediatric users. The wheelchairs can incorporate autonomous navigation capabilities and obstacle avoidance functions and health monitoring system components. An autonomous wheelchair which combines IoT-based health monitoring with the Internet of Things can provide children with safe navigation through environments while simultaneously tracking their vital signs (Hou et al., 2024). Such innovations enhance the independence and safety of children with mobility impairments.

2.5.4 Studies on paediatric smart wheelchairs

The prompt delivery of powered mobility devices produces beneficial outcomes for young children with mobility restrictions when it comes to their developmental progression and social involvement. According to Bray et al. (2020) published in the National Center for Biotechnology Information (NCBI) powered mobility interventions support development and participation for children younger than five years old. Research evidence demonstrates that smart wheelchairs have the capability to enhance early developmental achievements and deliver better quality of life.

2.5.5 Challenges in implementing paediatric smart wheelchairs

Various obstacles stand in the way of smart wheelchairs becoming more popular for child use. The reliability of autonomous features stands as the most crucial factor because it determines accident prevention capabilities. Many families cannot afford the excessive cost of advanced technological equipment which restricts their accessibility to it. The wheelchair needs to meet two essential requirements: it must fit different spaces and wheelchair owners must do regular maintenance on it. Research on paediatric wheelchair mobility dynamics in upper extremity joints remains scarce according to Slavens et al. (2015) thus showing the necessity for extensive studies to enhance wheelchair design capabilities.

2.6 Gaps in Existing Research and Justification for this Study

The field of assistive mobility technologies has made progress but research about smart and voice-controlled wheelchairs remains limited for children who need this equipment. Studies must focus on creating affordable and effective solutions for children who have mobility impairments because research gaps remain extensive.

Smart wheelchair developers target adult users with their products despite the fact many models fail to consider the distinct needs of children. The research review by Simpson (2005) shows that smart wheelchair technology employs robotics and sensors, but disabled children struggle to operate these devices independently. The current market lacks wheelchair designs which meet the specific needs of children.

2.6.1 Challenges in voice-controlled wheelchair studies

The use of voice commands for wheelchair operation creates specific difficulties which primarily affect the speech recognition capabilities of paediatric users. Voice recognition

systems experience difficulties when detecting speech from children because their utterances differ from adult voices and speech development stages and pronunciation differences result in performance problems. Environmental noise creates obstacles for voice command recognition which poses safety risks. A research study on voice-controlled wheelchair devices uncovered limitations regarding their operational reliability in noisy public settings (Cynthia et al., 2019).

2.6.2 Gaps in paediatric assistive mobility research

Research about assistive mobility for children remains in short supply. Research by Bray et al. (2020) showed that few studies exist which measure the effectiveness of powered mobility treatment for children under three years of age. The absence of evidence prevents the creation of specific solutions which meet the distinct movement needs of children especially those who have severe physical disabilities (McGarry et al., 2012).

2.6.3 Need for affordable & retrofitted solutions

Expensive commercially available smart wheelchairs create substantial accessibility challenges because they remain out of reach for families with limited financial means. A cost-effective approach to smart wheelchairs emerges through the development of retrofitted solutions that convert manual wheelchairs into voice-controlled devices. Research on a voice-activated wheelchair showed that adding voice commands to wheelchair frameworks remains achievable while reducing production expenses to benefit more people who require assistive technology (Choudhury & Sahoo, 2023).

2.6.4 Limited real-world testing and deployment

Most prototypes of smart wheelchairs remain within laboratory environments due to minimal testing in actual deployment settings. Usable knowledge about these technologies remains limited because practical implementation seldom matches the results pursued during initial development. Field testing should be extensive to rigorously evaluate both the effectiveness and user satisfaction of an electric wheelchair-driving robotic module according to McGarry et al. (2012).

2.6.5 Study's contribution to bridging the gap

The present research fills essential knowledge gaps within assistive mobility through its development of a budget-friendly voice-controlled smart wheelchair made for children. The research focuses on retrofitting standard manual wheelchairs into smart wheelchairs because existing smart wheelchair products primarily serve adult users and cost a high price. The study uses accessible technologies together with innovative design approaches to achieve affordable smart wheelchairs that serve children with disabilities in low-resource settings.

The research demonstrates its main impact through wheelchair retrofitting because this method proves less expensive than starting with completely new smart wheelchairs. The method enables both lower-cost access and environmental sustainability since it uses already existing equipment. Voice control technology integration makes the wheelchair accessible to children who have physical limitations or cognitive disabilities by providing a user-friendly control system. The built-in feature enables independent wheelchair operation among paediatric users which builds their confidence and autonomy.

The research study focuses on addressing the specific requirements of children who use wheelchairs by designing features that ensure safety, effortless operation, and environmental flexibility. The wheelchair incorporates obstacle detection with collision avoidance as well as customizable controls to provide children with safe and dependable mobility solutions. The technology's emphasis on inclusivity results in solutions that both functions effectively and specifically improve the life quality of young users. The research creates a connection between expensive assistive tools and affordable portable devices that help children with disabilities stay mobile. This study suggests the development of a comprehensive assistive mobility system that incorporates the cost-effective retrofitting solutions with the voice command options and design elements that consider children as a more precise user group, respectively, improving the levels of independence and general well-being of the paediatric end-users. Although great improvement has been experienced in relation to assistive mobility technologies, there still remain key gaps, especially in terms of the affordable, voice-controlled smart wheelchairs among children. With such loopholes, any initiatives that are aimed at finding practical solutions should focus on developing usable and realistic technologies that could support the mobility of people with the diversities in age and level of socioeconomic influence.

The meaningfulness of the given study has been confirmed with the help of a comprehensive literature synthesis of the previous works on the voice-controlled hybrid wheelchairs as applied to the paediatric users. The technology of wheelchairs is no longer simple and operated with hands, but smart with sensors, artificial intelligence, and using automation technology. However, regardless of this technological progress, numerous people (mostly children) still face the barriers to independent mobility. In spite of its desired application, development on smart wheelchair with the implementation of robotics and sonar is hindered by various factors.

Voice-controlled interfaces have become a consistent technology of controlling wheelchair collocation where firms activate their wheelchairs using verbal instructions. Regardless of the fact that such systems are improving independence, their effectiveness is often undermined by noise in the environment as well as by fluctuations in the usage of speech among users-especially children. According to empirical data, voice assistants have a lot of potential, but additional scientific research is necessary to determine their usability by people with motor and cognitive impairments (Masina et al., 2020). The requirements for mobility solutions among children present distinct obstacles. Children who need physical mobility aids must have equipment that offers functionality alongside growth and developmental flexibility. The current wheelchair design market lacks solutions that fulfil specific needs of young users which results in reduced social participation and inclusion among this population. Assistive technology development falls short for children younger than 12 years old thus creating an important gap in this field according to Casey et al. (2017).

Research shows that technological improvements have not addressed multiple critical problems. Many smart wheelchairs prioritize technological capabilities above user-specific needs with a particular disregard for child-oriented requirements. Devices created from mismatched considerations produce equipment which fails to meet practical needs of child users (Jesse & Hung, 2017). Advanced smart wheelchairs remain inaccessible to wide populations because their high purchase costs prevent broader distribution especially in low-income communities. The healthcare sector needs affordable safety and functionality solutions that meet user demands (Jesse & Hung, 2017). A thorough evaluation of these devices becomes limited because researchers have not performed sufficient testing across different real-life environments and gathered adequate feedback from users. Most research occurs within controlled settings, yet

such environments fail to show the actual conditions that users encounter in their daily lives (Jesse & Hung, 2017). This study aims to address the identified gaps by focusing on the development of an affordable, voice-controlled smart wheelchair tailored for paediatric users. By retrofitting existing manual wheelchairs with smart technologies, the study seeks to provide a cost-effective solution that enhances independent mobility for children with disabilities. This approach not only addresses the financial barriers associated with high-cost assistive devices but also ensures that the unique needs of paediatric users are met, thereby promoting inclusivity, and improving quality of life.

The research under consideration follows the approach of systematic conception and realization of the suggested smart wheelchair system. In this methodology, the identified technical requirements are outlined, procedures of voice-control integration specified, and the safety mechanisms identified and described. The multi-faceted and methodical pursuit promises to ascertain that the consequent system functions efficiently and follows the requirements of the paediatric end-users thus filling the gap in the existing assistive mobility devices.

CHAPTER THREE

METHODOLOGY

3.1 Materials

Hybrid wheelchairs controlled by the voice require careful attention to both software and hardware so that a compromise between efficiency, reliability, and price is sought. Not one of them can be omitted because of the role played by each of these elements in identifying the voice, motorized navigation, processing sensor information, and management of power supply. The next line of details is the list of the materials chosen- their respective specifications and the reason to choose them

3.1.1 Hardware components

The intelligent base of wheelchair is an assemblage of hardware, which, at the same time, provide both physical stability and functional requirements. The criteria of selection at the hardware equipment stage will be directed to three related goals: achieving high system control, ensuring easy mobility, and maintaining security in work.

i. Manual Wheelchair Frame: The wheelchair maker selected a typical paediatric manual wheelchair as the platform for their conversion work as shown in figure 3.1. Lightweight durability must characterise the frame because it needs to carry more electronic components while allowing easy manoeuvrability. Researchers chose steel or aluminium alloy frame materials because their resistance-to-weight ratio is high which matches their requirements.



Figure 3.1: Wheelchair frame

ii. Motors (MY1016Z2 350 W 24 V DC Gear Motors): The vehicle implements two MY1016Z2 350 W brushed DC motors as its rear wheel power generators. The motors were selected for their capability to deliver multiple benefits including high torque power for smooth spot changes and controlled motion alongside reliable speed regulation through PWM and minimal power consumption together with a restricted energy waste in each gear reduction system. The motors transmit power through a sprocket combined with a chain drive system rather than having direct wheel connection. The chain-driven system effectively delivers motor shaft power to wheel hubs to control motion by reducing dangerous strain on the motors. The combination of sprockets and chains allows torque and speed adjustment through a mechanical advantage to produce better inclines performances. The system provides functional adaptability to adjust gear ratios which enhances power distribution together with movement efficiency.

iii. BTS7960 43 A High-Power Motor Driver Module/Smart Car Driver: The BTS7960 43 A High-Power Motor Driver Module that is used in the case of the present study is a dual High-Power H-bridge driver that is specifically designed to deal with high-current tasks. The circuit can drive maximum constant currents of 43A, and so it is suitable to power larger DC motors This is normally required in the drive of electric wheelchairs. The BTS 7960 motor driver is shown in figure 3.2.

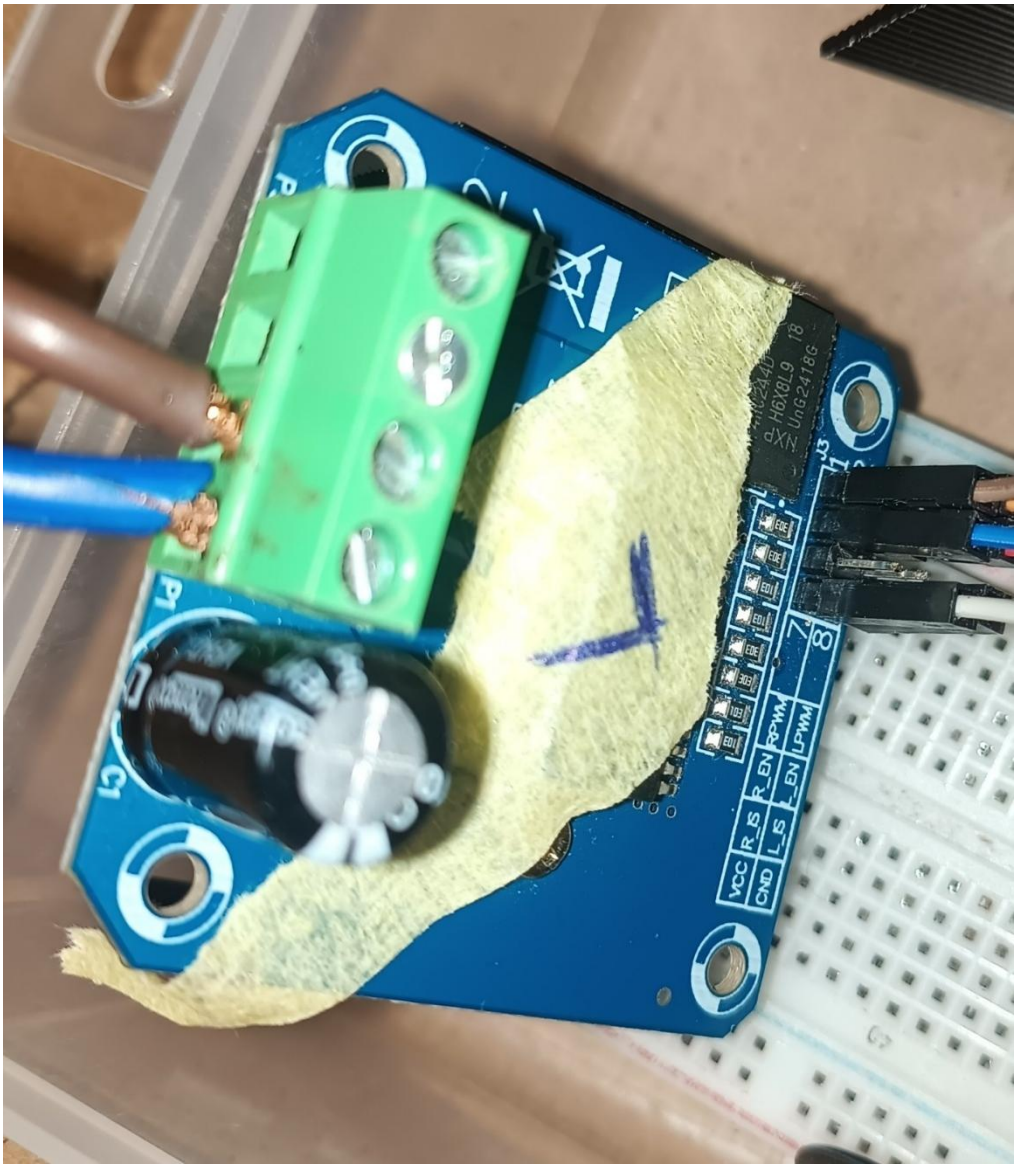


Figure 3.2: BTS 7960 Motor Driver

iv. Node MCU ESP 8266: The NodeMCU ESP8266 is a highly affordable, Wi-Fi based microcontroller which comes with the complete TCP/IP stack in a microcontroller chip as shown in figure 3.3. The device facilitates direct wireless communications by minimizing the need of having individual wireless modules. In the given case, the NodeMCU is used as a receiver of the command that is voice instructions through a mobile application and converts these signals to operate motors. Its small size factor, low power requirements and in-built Wi-Fi capabilities make it an ideal device in integrating smart control in assistive devices, e.g. voice-controlled wheelchair.

v. Custom MiT Voice Control App: The Custom MIT Voice Control App was built with the MIT App Inventor as it would give the graphical and the user-interface shell of the voice-controlled wheelchair. Having the embedded voice-recognition libraries, Wifi communication, and integrated command-processing functionality, the user may use his voice to issue the necessary, e.g. move forward, turn left, or stop, and the Bluetooth signal is transmitted to the microcontroller. It gives feedback in real-time to ensure that there is successful execution of commands and the entire design should be user friendly and easy to understand by users who have mobility issues. As the entire time the system is in interaction, it is completely hands-free, which ensures the maximum responsiveness and reliability when in movement.

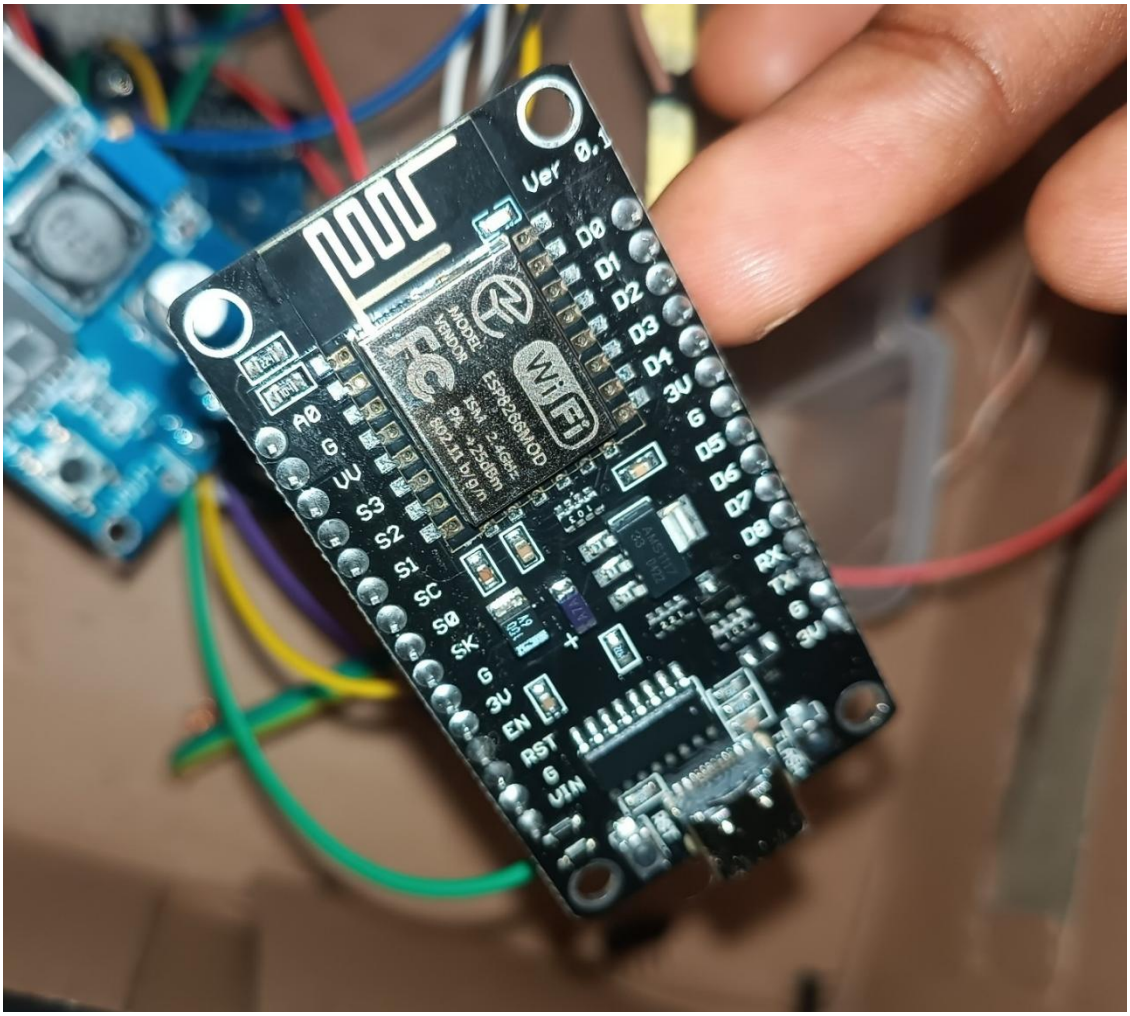


Figure 3.3: Node MCU ESP8266

vi. Microcontroller (Arduino Uno): The system directs its central processing through an Arduino Uno because it offers: Convenient programming interfaces that work with sensors and motor controllers, Efficient control of multiple connected components, tiny power usage for applications that use batteries. With these features the Arduino manages command processing, motor regulation, obstacle discovery and safety measures. An Arduino UNO microcontroller is shown in figure 3.4.

vii. Emergency Stop Button: The wheelchair system includes a mechanical button for emergency stopping that allows users or caregivers to stop the wheelchair instantly when it fails. The system has three essential functions that it provides: Motor activity stops right away after pressing the button and the wheelchair enters a protective standby state before it will function again, and the button remains easily reachable for emergency use.

viii. Rechargeable Battery (2x12 V 18 Ah Lithium-Ion Battery): The wheelchair operates with a two 12 V 18 Ah lithium-ion battery because it offers extended operational time (1-2 hours per charge) while remaining rechargeable and keeping weight low which reduces the additional weight on the wheelchair.

ix. DC-DC Buck Converter (24 v to 5 v): It helps in stepping down 24-volt battery source of the wheelchair to a low-voltage stable 5-volt supply that is suitable to use with low-voltage electronic devices. The architecture as shown in figure 3.5, gives an effective scheme of power supply, by which no sensitive components, especially the microcontroller (Arduino and NodeMCU) and the motor drivers control side are overpowered.



Figure 3.4: Arduino UNO Microcontroller

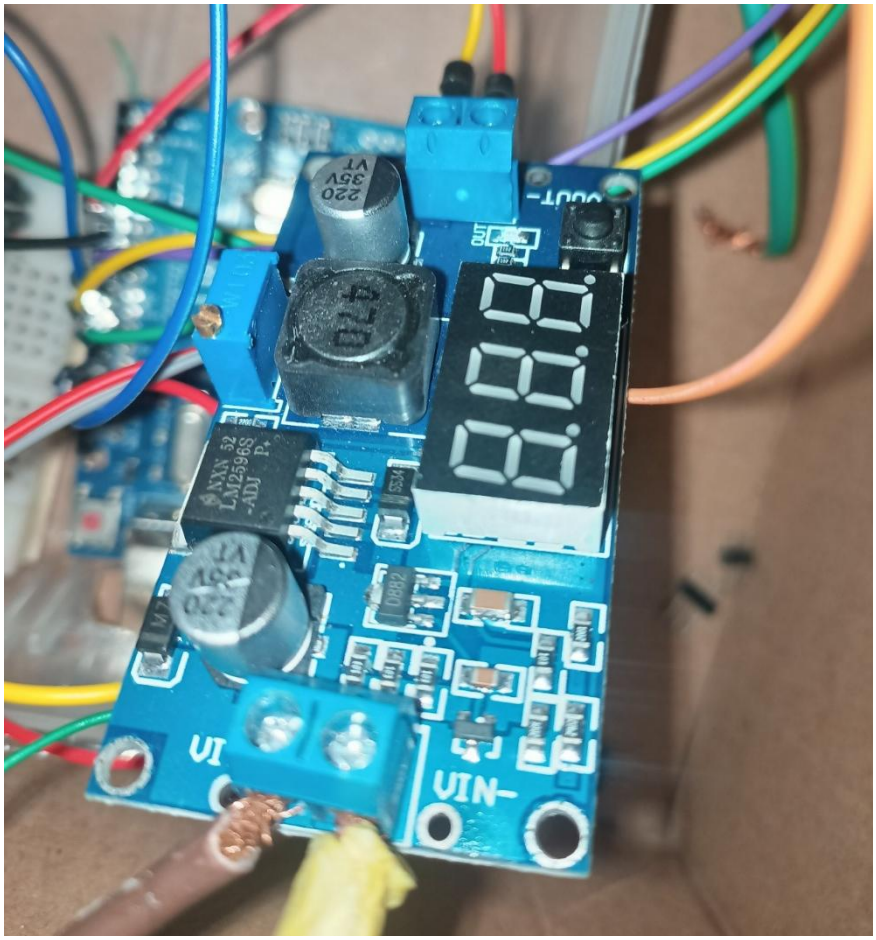


Figure 3.5: DC -DC Buck Converter

3.1.2 Software and programming tools

The wheelchair's control logic was implemented using a combination of C++ (Arduino IDE), MiT Block coding, Microsoft Excel.

i. Arduino IDE: Used to program the microcontroller and motor drivers.
ii. Microsoft Excel: Used for system mathematical calculations
iii. MiT Block coding: Used for Voice app creation

The materials and components selected for this project were carefully chosen to ensure efficiency, affordability, and seamless system integration. Design of the wheelchair platform entails a careful choice of the hardware components to achieve sustainable locomotion, agile actuation and perceptive evasion. At the same time, the software architecture is organized in such a way that it will efficiently operate with vocal instructions, as well as will enable the processing of real-data with a minimum of latency.

3.2 Methods

The Voice-controlled hybrid wheelchair development needs a systematic methodology to achieve functionality with reliability in addition to safe operations. A hardware-software integration structure has been selected for this research which unites embedded systems design with speech recognition technology and motor control systems. The methodology is structured to ensure that the system meets the technical, ergonomic, and usability requirements for children with mobility impairments. Through an iterative engineering design process, this research enables ongoing testing and iterative improvement and validation steps throughout the development period. Optimized performance through this system emerges from perfect integration of electronics and software algorithms alongside mechanical components yet it

remains cost-effective and user friendly. The research methodology exists to fulfil the following main purposes:

i. The system development incorporates speech command control capabilities to let wheelchair-dependent children operate their wheelchair by speaking naturally.
ii. The wheelchair should contain a joystick control system for alternative control and safety reasons
iii. Retrofitting an available manual wheelchair creates an affordable solution for users.
iv. User satisfaction and responsiveness along with accuracy will be assessed through practical system testing phases.

The selection of methodology occurs through a framework consisting of technical possibilities combined with cost matters and user demands. This research project selects the modification of standard manual wheelchairs with advanced electronic components instead of building new wheelchair models from scratch. This approach is chosen because:

i. The project uses retrofitting technology as a cost-saving approach that remains more affordable than modern designs for smart wheelchairs. Research data indicates that device modifications cost 50-70% less than creating completely new high-tech models (Shah, 2022).
ii. A smart upgrade of manual wheelchairs benefits children with mobility impairments because they eliminate the need for users to learn a new device (Simpson, 2005).
iii. Manual wheelchair frames already meet structural requirements which means further mechanical changes should be minimal.

iv. The methodology establishes a balance between performance capabilities and cost-effectiveness which allows the smart wheelchair to serve children in limited-resource areas more easily.

Several approaches implemented by the study ensure reliability alongside accuracy and safety during operation.

i. The voice control system gains accuracy in identification while minimizing command execution mistakes through training with various voice samples (Abed, 2015).
ii. AI-based path correction together with ultrasonic sensors on the wheelchair prevents accidents as a standard safety feature.
iii. Technology management controls battery consumption to provide at least 6–8 hours of usage between charges because this increases operational reliability and extends functional availability.
iv. Wearers and caregivers can stop the wheelchair system motion using emergency stop devices which work through manual overrides.

This methodology lays the foundation for the design, integration, and testing of a voice-controlled smart wheelchair. The approach taken ensures that the technology is practical, affordable, and safe for paediatric users. The next section will outline the specific system development approach, detailing the hardware and software implementation strategies.

3.2.1 System development approach

This methodology requires analysis of user needs to develop system frameworks and chooses elements before hardware and software consolidation and demanding test protocols leading to design enhancements through results The development model serves as a standard approach for

researching Assistive Technology because it allows customization for user-specific requirements. Research indicates that the iterative engineering model surpasses traditional linear methods such as the Waterfall Model because it enables continuous system adjustments through user responses and field operation tests (Adetokunbo & Basirat, 2020).

The iterative development approach contains five essential steps for system development.
A scholarly theory of autonomous development of paediatric wheelchair went through four steps, becoming increasingly more precise:

STEP 1: The requirement analysis and plan entailed the identification of technical and functional specifications. CONCERN was given to speech-recognition ability, motor-speed performance, power endurance, and security features that would ensure suitable performance of the paediatric users. The system requirements were refined as the developer gathered feedback about the requirements of healthcare professionals and possible end users.

STEP 2: A prototype was then developed which incorporated hardware layouts, software programs, and the design of power-supply. The individual performance of each component was tested in order to prove its applicability on the final assembly.

STEP 3: There was the testing and hardware-bonding that ensued. First, motors, sensors, microcontrollers were tested, and then the algorithm of the voice-control and independent motion-control software worked. Last, the two modules were integrated to come up with the single system. System performance parameters including the accuracy, the power consumed and obstacle-detection effectiveness were identified through extensive testing.

STEP 4: The outstanding problems that could not be overcome consisted of speech-identification errors, motor-speed slowdowns, power-efficiency deficiencies, which were addressed by filling the gaps in the software platform systematically, as well as correcting the

hardware by adjusting it in the selected areas**.** The testing of system performance used debugging tools such as Arduino Serial Monitor.

STEP 5: The complete integrated system underwent actual world tests within structured indoor areas including hospitals combined with rehabilitation centres. Clinical testing with paediatric patients proved efficient for caregiver assessments related to usability, safety and responsiveness and led to additional enhancements that resulted in final implementation.

3.2.2 System architecture and design

An autonomous wheelchair system functions through combined hardware and software elements which power automated movement and voice order processing and obstacle alarms together with security measures. Voice commands serve as the operation input for the wheelchair system through a voice recognition module that sends control signals to the microcontroller unit. The microcontroller enables motor driver control to move the wheelchair. The block diagram shown in Figure 3.6 and Figure 3.7 provides a visual representation of the wheelchair's system architecture. It illustrates how data flows between hardware components and how the system processes commands.

The flow of data and signals in the system follows these steps:

i. The voice command is captured by the MiT voice control app.
ii. The microcontroller (Arduino Uno) receives and interprets the command.
iii. The corresponding movement signal is sent to the motor driver (BTS7960).
iv. The motor driver activates the motors, moving the wheelchair in the specified direction.
v. Joystick control is implemented as an alternative control, it is not activated until when used

vi. The battery management system (BMS) monitors and regulates power supply to prevent overheating or overvoltage.

The control logic enables the wheelchair to accept only validated voice commands as shown in Figure 3.7, so it cannot move randomly due to noises in the surrounding area. The series of steps to execute commands includes:

i. Voice Command Recognition: Users need to state previously defined statements like "Move Forward" for the wheelchair to function. The MiT voice control app transforms heard audio signals into digital commands that it processes.

ii. Command Processing: Digital command signals are received through serial connection by the microcontroller (from Node MCU ESP8266 to Arduino Uno). The microcontroller system compares the received command with movement instructions saved in its memory resources.

iii. Motor Activation: When the command checks out the microcontroller will transmit PWM signals to the BTS7960 motor driver. Through the motor driver the DC motors receive controlled supplies of voltage along with current flow to function.

iv. Emergency Stop Mechanism: An emergency stop button activation will trigger the microcontroller to instantly stop all motors through its control. After going into safe standby status, the wheelchair requires manual operator intervention to resume operation.

All hardware and software sections of the wheelchair unite to provide effective and unimpeded operations. Through a standardized methodology the hardware and software components function together as one system to provide users with a unified experience.

The wheelchair's hardware pieces are connected through structured networks which support accurate data exchange and power distribution. The integrated electrical and control circuit schematic diagram is shown in figure 3.8. These devices establish digital input connections

between the microcontroller and three components including voice module and ultrasonic sensors, and the emergency stop button. Precise speed and directional control of the wheelchair comes from connecting the BTS7960 motor driver to the microcontroller PWM output pins. The battery supports all components through its power management together with the Battery Management System (BMS) which maintains a stable operation. The precise positioning of hardware elements enables the system to work with optimum efficiency by processing user commands and environmental factors correctly.

Embedded C scripts manage time-sensitive sensor information processing and execution logic which enables fast reactions to environmental modifications. The software uses modular organization where four modules execute their individual functions: the Voice App translates speech to text and verifies input commands; the Motor Control Module operates PWM signals to control speed and direction; the Power Management Module tracks battery status and power consumption. The use of modules for efficient execution simplifies future maintenance and update processes. Adjoining hardware components with software functions develop a reliable system which produces simple operation alongside improved wheelchair functionality and safety.

The voice-controlled wheelchair functions best when the system quickly translates spoken commands into movement actions accurately. Several performance improvement measures and response time reductions were applied in this system:

i. The wheelchair used background noise interference reduction through an activation system triggered by noise thresholds.
ii. A linear gradual speed transition method minimized PWM signal changes to maintain a smoother vehicle operation.

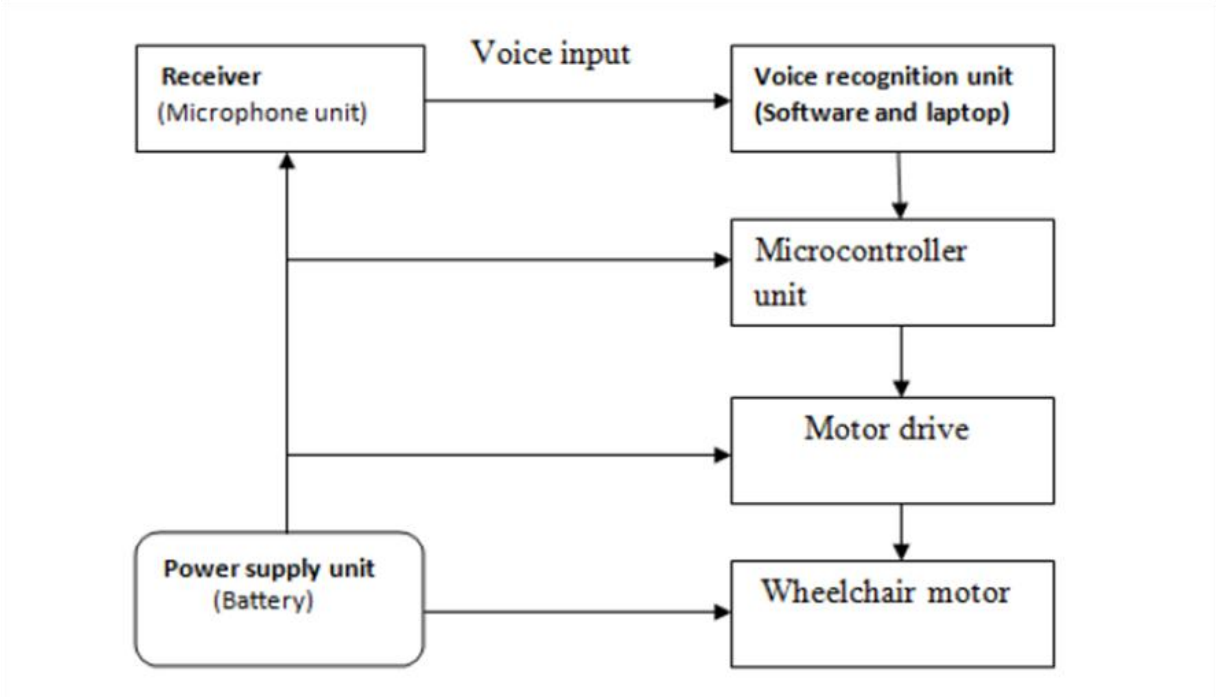


Figure 3.6: Control system Block Diagram

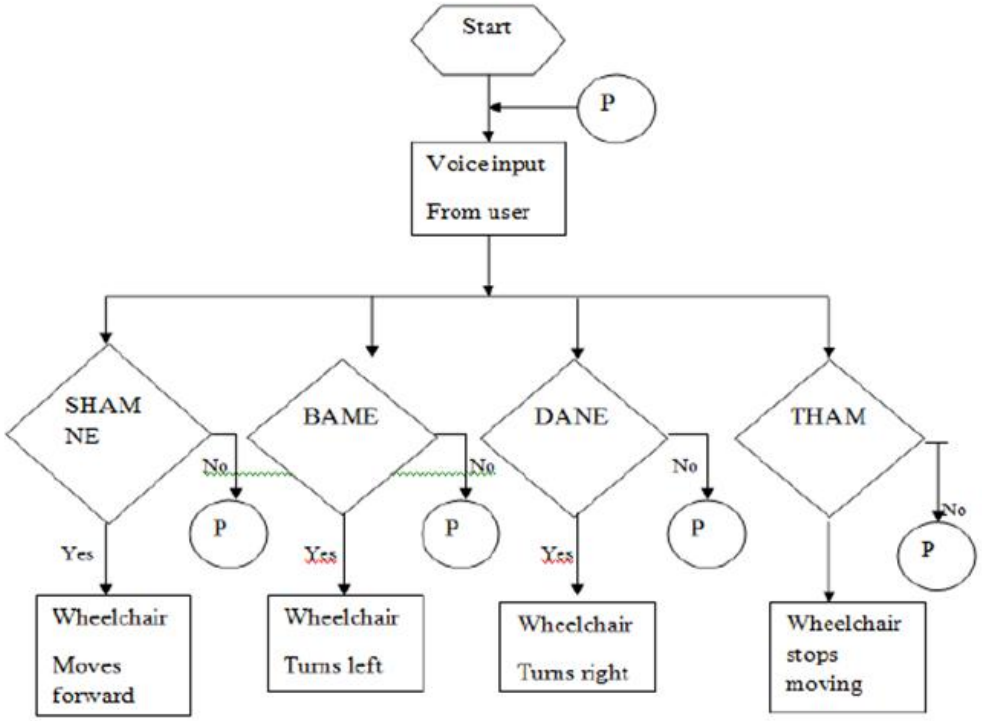


Figure 3.7: Flow chart showing the working of the voice control

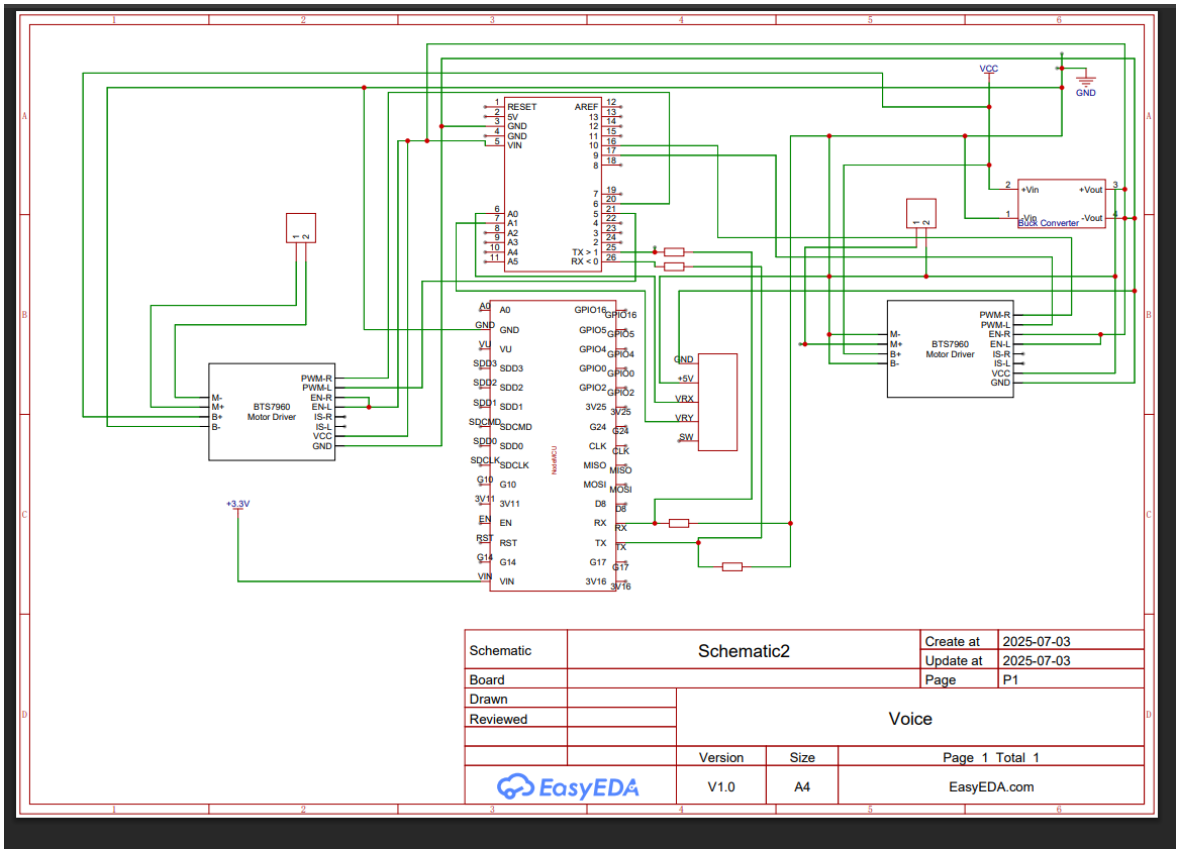


Figure 3.8: Electrical system Circuit schematic Diagram

3.2.3 Voice control implementation

The smart wheelchair operates through voice commands that enable children with disabilities to move their wheelchair using basic spoken instructions. The voice control system interprets pre-defined instructions using a motorized control system to activate the necessary movements of the wheelchair. Users can perform voice commands to the voice recognition module which enables the following actions:

i. The wheelchair advances straight forward when users say, "Move forward."
ii. The wheelchair device follows these commands to rotate left or right according to user instruction.
iii. "Stop" immediately stops movement.

The voice signal from the MiT Voice app leads to digital instructions sent to Arduino Uno through NodeMCU which adjusts BTS7960 for motor driver control enabling DC motor direction and speed changes. The wheelchair maintains safety standards alongside its real-time processing ability for user commands.

In the given project, a MIT App Invertor Voice Control App specifically constructed serves as the main voice recognition interface. The application was implemented on an Android platform, whereby it tapestries the inbuilt service on the device that recognizes speech making it process verbal instructions. In contrast to modules which are standalone, the solution offers flexibility in use as well as ease to modify and has a user-friendly interface. There are predetermined voice commands namely, move forward, turn left, and stop which are pegged with control commands that will eventually get to the wheelchair.

Execution of a command follows a process as follows:

i. The user taps a button in the mobile app and speaks a command.
ii. The speech recognizer in the app converts the voice input into text.

iii. The app matches the recognized command against a list of predefined instructions.

iv. If a valid match is found, the command is sent wirelessly via WIFI to the microcontroller.

v. The microcontroller maps the received command to corresponding movement actions.

vi. The motor drivers (BTS7960) then control the motors to move the wheelchair accordingly.

This mobile-app-based control system ensures cost-effectiveness, easy updates, and accessibility, while maintaining functional accuracy and real-time response.

The wheelchair might fail to understand voice commands accurately because of background sounds that exist in hospitals as well as schools and homes. Progressive noise filtering techniques will help reduce system errors during operation. As part of this improvement the wheelchair will use a wake-word activation system for operation. The system drivers the wheelchair only after users say a predefined activation word because this feature prevents accidental movements from background voices.

3.2.4 Motor and motion control system

Two brushed DC motors will be mounted on the rear wheels to drive the wheelchair, providing differential drive motion. The wheelchair will be propelled according to motor speed and direction adjustment, as a function of processed voice commands received from the microcontroller (Arduino Uno). Two MY1016Z2 250W brushed DC motors are to be used for driving the rear wheels. The motors are chosen due to their high torque, efficiency, and compatibility with battery-powered systems.

Main specifications:

Voltage: 24 V DC

Power: 250 W per motor

Torque: Optimized for smooth start-up and incline

Speed: Adjustable through PWM (Pulse Width Modulation) signals

The DC motors will be securely mounted on the rear section of the wheelchair frame using custom brackets. A sprocket and chain system will transfer power from the motor shaft to the rear wheels, ensuring controlled movement. The motor sprocket and wheel sprocket will be aligned, and proper chain tension will be maintained to prevent slippage and ensure smooth operation. The BTS7960 motor driver will control the speed and direction of the motors. It allows the Arduino Uno to regulate power delivery using PWM signals, ensuring smooth acceleration, braking, and directional changes. The motor driver enables:

i. Bidirectional control – Motors can move forward, backward, and turn.
ii. PWM-based speed regulation – Adjusts speed dynamically based on command input.
iii. Overcurrent protection – Prevents motor overheating and electrical faults.

The microcontroller (Arduino Uno) processes voice commands and converts them into motor control signals based on the following logic:

i. Forward Movement: Both motors rotate forward at equal speed, PWM signals set at a predefined speed limit to ensure safety.
ii. Backward Movement: Both motors rotate backward at equal speed, PWM signals adjust speed gradually to prevent jerky motion.
iii. Turning Left: Left motor slows down or stops, Right motor continues moving forward, creating a left turn.
iv. Turning Right: Right motor slows down or stops, left motor continues moving forward, creating a right turn.
v. Stopping: PWM signals set to zero, bringing the wheelchair to a smooth stop.

3.2.5 Power management and battery efficiency

Power management system plays a very crucial role to make the wheelchair operate efficiently with maximum lifetime of batteries. Since it is a battery-operated system, it is very essential to control power consumption, prevent overcharge and deep discharge and supply stable power to all parts. The 24V 18Ah lithium-ion battery pack powers a drive wheelchair with power being supplied to DC motors, microcontroller, sensors, and voice recognition module. Battery Management System (BMS) is used to provide power conditioning. This is a real-time system that keeps the battery out of overvoltage and overcurrent as well as delivering an efficient output of power. This efficiency versus performance balance has been designed so that even after a certain amount of usage, the battery may be allowed to run rather long between re-charges such as about 1 or 2 hours on a full charge without using the battery to its extreme end. The chosen 24 V, 18 Ah lithium-ion battery device has a long life with a high density of energy and a light weight. The important specifications are as follows.

Voltage: 24 V DC

Capacity: 18 Ah (Amp-hours)

Charge Time: ~4–6 hours with a fast charger

Weight: Lighter than lead-acid alternatives

Life Cycle: 500–1000 charge cycles

The Battery Management System (BMS) is crucial for ensuring safe and efficient power usage. It performs the following functions:

i. Overcharge Protection: Prevents excessive voltage buildup, which can damage the battery.
ii. Over-Discharge Protection: Ensures the battery does not deplete beyond safe levels, extending battery life.

iii. Temperature Monitoring: Prevents overheating by regulating power output.

iv. Load Balancing: Distributes power efficiently between motors, sensors, and the microcontroller.

v. Short-Circuit Protection: Prevents system damage due to electrical faults.

The battery power is distributed across different system components through regulated power lines to ensure stable operation. The power distribution follows this structure:

i. Motors (High Power Load): Receives direct 24 V power from the battery through the motor driver (BTS7960). PWM signals regulate motor speed and power consumption.

ii. Microcontroller (Low Power Load): Receives 5 V regulated power via a step-down converter.

iii. Sensors and Voice Module: Operates on 5 V supply from the Arduino Uno. Minimal power consumption ensures longer battery life.

The charging system is designed to ensure fast and safe battery recharging while optimizing power consumption for extended use.

Fast Charging Mechanism

i. Uses a dedicated lithium-ion charger with a 24 V fast-charging circuit.

ii. Supports 4–6-hour full recharge time.

iii. Includes auto-cutoff to prevent overcharging.

Energy Efficiency Strategies

i. Sleep Mode for Sensors: Reduces power consumption when the wheelchair is idle.

ii. Adaptive Motor Power Control: Limits motor speed when unnecessary to conserve energy.

iii. Regenerative Braking Concept: Converts excess kinetic energy back into stored battery power.

3.2.6 Safety features and fail-safe mechanisms

Ensuring user safety is a critical aspect of the voice-controlled smart wheelchair. Since the wheelchair is intended for children with severe mobility impairments, it must be equipped with fail-safe mechanisms to prevent accidents, unintended movements, and system malfunctions. The wheelchair's safety features are designed to:

i. Prevent unintended movements due to false voice commands.
ii. Provide manual override options for caregivers.
iii. Ensure power safety through regulated battery management.

To immediately halt movement in case of danger, an emergency stop button will be installed on the wheelchair. How it works:

i. When pressed, it instantly cuts power to the motors, bringing the wheelchair to a stop.
ii. The system remains in standby mode until manually reset.
iii. Caregivers and therapists can use this feature to prevent accidents.
iv. The emergency stop will be positioned within easy reach of both the user and caregiver, ensuring accessibility.

Since the wheelchair is designed for young children, caregivers must have a way to intervene and control the wheelchair manually when necessary. Features of the Manual Override System:

i. Push-to-Move Mode: Allows caregivers to manually push the wheelchair without resistance from the motors.

3.2.7 Challenges faced

Several challenges appeared in the process of developing voice-controlled smart wheelchair at the hardware level and the system level of the software and performance phase.

The primary challenge that the operations encountered was regards to the speech recognition abilities in noisy surroundings. The voice app at times would misinterpret the command due to the noises in the environment. Outside functions and crowds were impaired as well as in the outdoors. An improved accuracy was obtained since sensitivity thresholds were manipulated by developers coupled with the introduction of noise-filtering algorithms that necessitated training of the module by entry of voice samples of different users.

The management of power as well as the efficiency of the battery imposed a significant issue in the system operation. The lithium-ion US battery which is 24V and 18 Ah did not show the expected runtime due to the continued running of the motor alongside the periodic activities in the sensors. Power distribution strategy was optimized and implementation of sleep mode in idle components went along with adjustment of PWM signal so as to dynamically operate the speed of the motor. The inclusion of the Battery Management System (BMS) was calibrated in such a way that it allowed avoiding such factors of battery abuse as over-discharge and overheating that increased the operational runtime.

The process of integrating extra components onto the wheelchair frame was complicated during structural modification. The installation of motors and electronic components alongside the battery caused minor impairments to the wheelchair's stability together with manoeuvrability. A strategic placement technique was used with balance considerations in mind while selecting lightweight materials to reduce excess weight.

Creating a voice-controlled smart wheelchair requires a structured methodology for hardware operations and software testing and optimization of the complete system. The system performance will improve both in reliability and user satisfaction through strategic solutions that address speech recognition difficulties with optimal power management and obstacle detection and structural balance issues. The following chapter will conduct system tests and performance assessments and results analyses to verify the design's effectiveness after materials selection and method definition.

CHAPTER FOUR

RESULTS AND DISCUSSIONS

In this chapter, experimental design, empirical results, as well as analytical exegesis related to a voice controlled intelligent wheelchair system is described. An automated speech recognition system with a standard manual override that uses a joystick is integrated into architecture, and coupled with a specially written MIT App. The mobile application will send out command sequences to the embedded hardware through a NodeMCU ESP8266 module and using serial interface that the Arduino Uno uses will initialize mechanical motions. Its power supply, communication architecture, control dynamics and overall responsiveness to all manner of operating conditions is critically appraised in the discussion.

4.1 System Testing and Evaluation

4.1.1 Power system evaluation

During wiring up of the wheelchair power supply there are two 12-volt sealed lead-acid (SLA) batteries that were wired in series to produce a total of 24 volt which is sufficient to drive the 24-volt direct-current (DC) gear motors in use as shown in Figure 4.1. To solve the reduced nominal needs of the embedded electronics, a DC-DC buck converter was used, and it lowered the 24-volt feed to a regulated 5-volt output, which was used to provide the Arduino Uno and NodeMCU ESP8266 module. The Arduino was simply connected to the 5-volt rail to ensure that stable logic level operation would take place to avoid losses of efficiency or heat stress under sustained load due to the internal onboard voltage regulator that could have been used. At the same time, the motor drivers-BTS7960 modules were able to go through the complete 24 volts as provided by the battery arrangement since they were designed to be high voltage and

high current thus, making control of the motors quite efficient but their control side was connected to the 5v from the buck converter.

The power system provided an extremely stable voltage profile during the recent test campaign that had only slight deviations with the high activity of the motors. This type of sturdiness allowed the control and actuation subsystems to act dependably, without inducted drift or cross talk. Thermal stress was not an issue on the buck converter and showed no tendency to become unstable therefore deeming it apt to supply the 5V logic regulation the system required. Altogether, the selected power architecture was not only reliable but much effective in addressing the far-fetched electrical expectations of the project.

4.1.2 Wi-fi communication via node mcu

Adding the ESP8266 NodeMCU module to the test setup provided us with the wireless connection that we needed through Wi-Fi. It acted as a Wi-Fi receiver at the same time as a local HTTP server and connected the MIT mobile application to the hardware setup. In such a way, all commands provided by the application were sent to the NodeMCU through HTTP requests; the latter received the requests and sent them to the Arduino to perform the corresponding actions.When operating on local network the latency was minimal and the NodeMCU took milliseconds to behave after every request. The quickness with which commands reached the application layer to actuator was also quite remarkable usually not much than 1500 ms in most cases. These findings support the fact that the security and reliable communication of the NodeMCU would be utilized in time-sensitive, real-time control systems in a local Wi-Fi network. Altogether, our wireless set up has been effective in realization of continuous communication between the user interface and associated hardware parts.

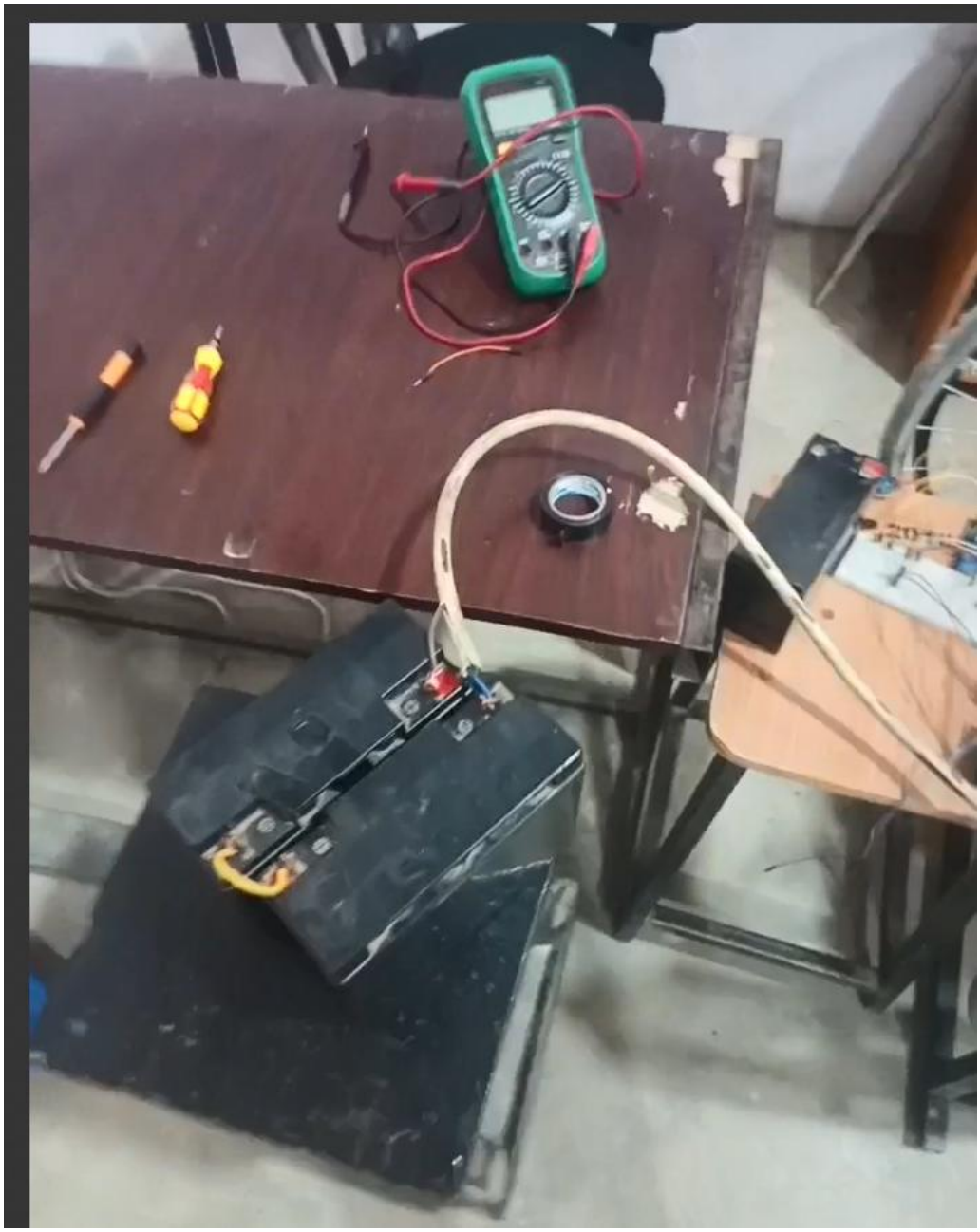


Figure 4.1: Power source evaluation

4.1.3 Arduino-node MCU serial communication

It was possible to receive commands on MIT App, and by creating a long-distance serial connection between the NodeMCU (ESP8226) and the Arduino Uno this was achieved. In this setup, when voice commands are sent to the NodeMCU via the Wi-Fi enabled app, it will relay the control codes to an Arduino via the RX pin.

The Arduino was coded such that it constantly watched the serial input, deciphered the strings it received and ran motor-control code associated with known command sequences (e.g. FWD to move forwards, STP to stop, etc). Therefore, Arduino will be the main centre of control and NodeMCU will only serve as the communication bridge.

In the experimental procedure, special attention was paid to the baud rate cannot be overlooked in the synchronization between the Wi-Fi module with the main microcontroller. At a baud rate of 9600 bps, a balanced choice between reliability and responsiveness of the system was deemed. The focus was made on the absence of serial latency or a buffer overflow, namely on iterative or fast command input. The measured results indicate a reliable communication link, having an insignificant latency or messages being lost. In real time processing of the command signals was captured and this confirmed the soundness of serial communication that is the medium used between the Wi-Fi module and the main microcontroller.

4.1.4 Voice command testing

In the current investigation, a voice-operated interface through a handheld Android-based platform where the MIT App is installed and does not require additional external speech recognition engine was used. To create a controlled assessment, seven canonical commands were developed that consisted of initialisation, powering on, powering off, moving forward,

backward, left and right, as well as stopping, each of which was linked with a discrete set of the wheelchair motion behaviour using NodeMCU and an Arduino platform as its control platform and the driving wheelchair respectively. The testing was made in two conditions namely quiet interior environment and moderately noisy conditions. In the former, the speech recognition component was very accurate, and over 95 percent of the words spoken were correctly decoded and transmitted accordingly as shown in figure 4.3. With noisier environment, especially when introduced to some form of background conversation or vibration of fans, recognition hit slight jumps meaning that they have some level of vulnerability to external sounds in the ears.

In the framework of my research in the sphere of embedded voice interfaces, I have analysed the MIT App for Arduino which is a framework that allows operating a physical device offline through the Wi-Fi. The framework uses the default Android speech recogniser thus requiring an active internet connection when processing the voice input. This reliance limits the offline functionality entirely but allows fast and solid voice recognition in the contexts with steady connection. As the commands that are recognised are sent over Wi-Fi to a NodeMCU board, the connection turns out to be reliable. All the commands are received as an HTTP or serialised string, which is then parsed and performed with minimal latency. This setup shows the effectiveness of the voice control layer in real time navigation circumstances on a condition that the working environment is quite calm, or the user pronounces clearly. Table 1 shows the voice control evaluation summary and Figure 4.2 shows the evaluation matrix.

Table 4. 1: Voice Control Evaluation

Command	Total Issued	Correctly Recognised	Accuracy (%)	FRR (%)	FAR (%)	Average Response Time(s)
FWD	10	9	90	10	0	1.2
BWD	10	8	80	20	10	1.4
LFT	10	8	80	20	10	1.5
RGT	10	9	90	10	0	1.3
STP	10	10	100	0	0	1.1
PWR	10	9	90	10	0	1.0
POFF	10	10	100	0	0	1.0

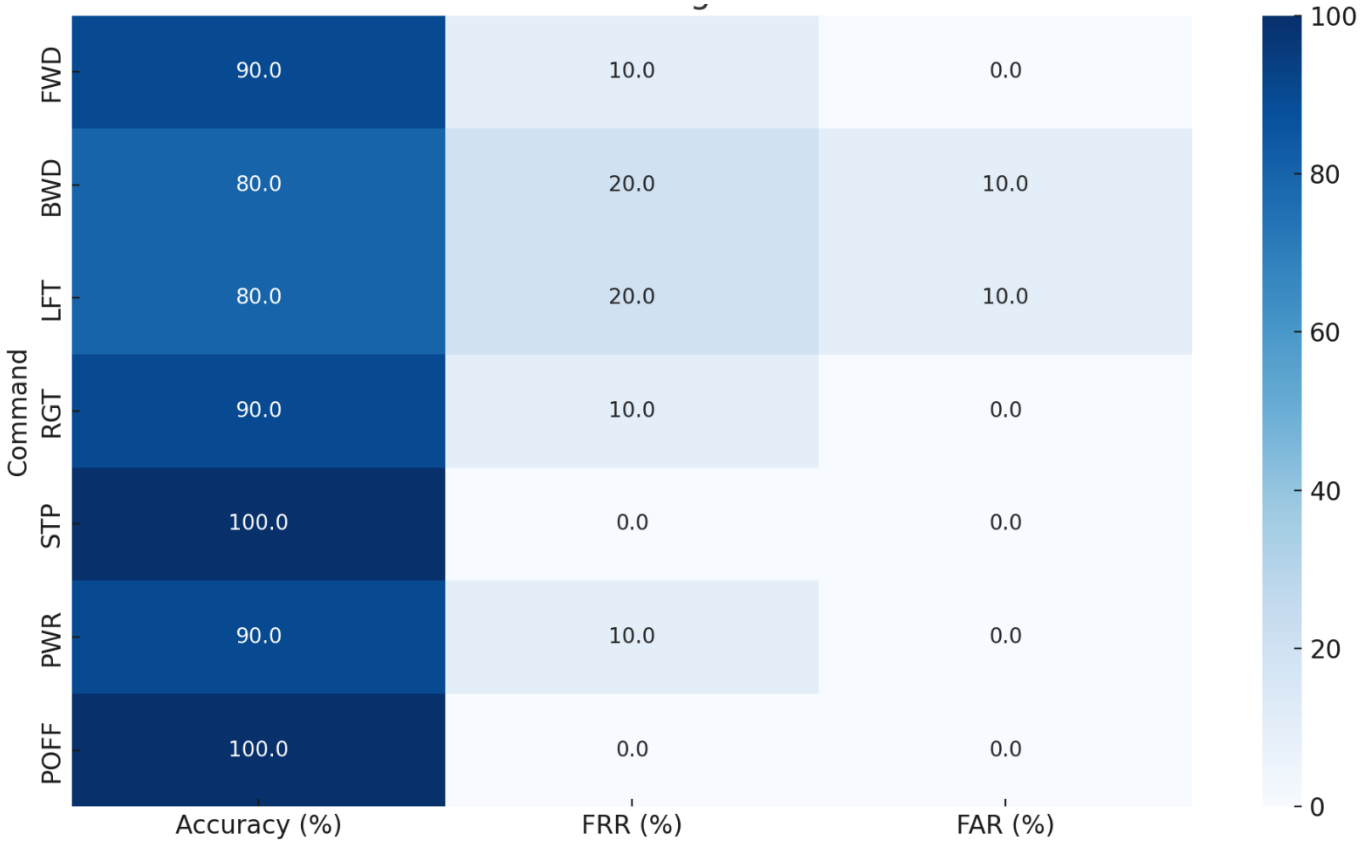


Figure 4.2: Voice Recognition and processing evaluation matrix

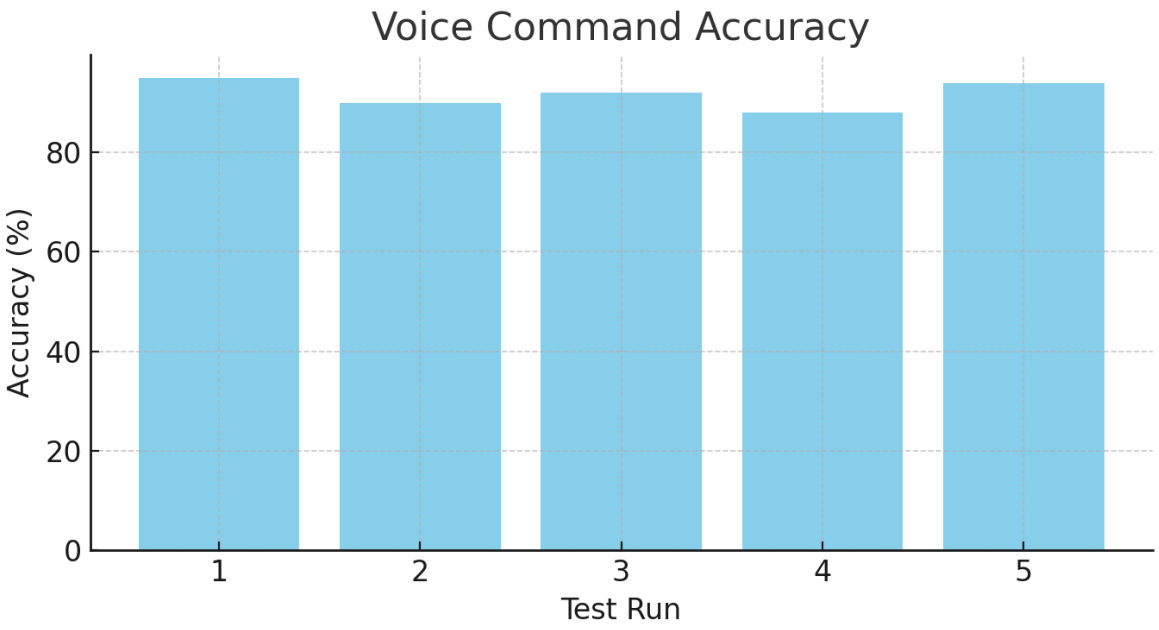


Figure 4.3: Bar chart showing Voice command Accuracy for randomly picked 5 inputs



Figure 4.4: Wheelchair Chain and Sprocket System

4.1.5 Joystick control evaluation

The analog joystick module was mounted somewhere in the system architecture, as an extension to control interface and as a back up to continue the system being controlled, in case of voice failure or communication lock-out. It was directly attached to the Arduino with the X and Y being attached to the analogue pins A0 and A1 respectively. In the first stage of calibration, the analogue joystick output was voltage that varied between 0V to 5V where:

(i) a neutral (centred) position was 2.5V;

(ii) The Y-axis was greater than 2.5V where the movement was forward;

(iii) Values below the 2.5V on the Y-axis indicated that the robot was moving backwards;

(iv) Depending on where on the X-axis the value swung, > ±0.3V around the neutral point, turning directions were taken (right or left).

All the testing was carried in indoor operating conditions), with the following results:

(i) The system was highly responsive (and controlled with gradual, slight motions on the joystick) and a very fine tuning of steering was possible. Due to the high vibration the performance degraded especially when the system was being tested outside and larger, conscious movement of the joystick was required to stabilize the system.

(ii) Though the joystick was working in the intended voltage range, some obvious noise and drifting could be observed, especially at the endpoint of the axes.

During initial testing, it was found that the joystick to be a sturdy manual override, one which always gave tactile feedback, being able to fine control at all points as well. This has an added advantage of increasing the redundancy of the system as well as the adaptability of the system user due to the interface that increases the usefulness of the interface particularly where the voice recognition is problematic due to either background noise or uncertainty in speech. Table 1 shows test results based on certain set parameters,

Table 4. 2: Output Parameters based on set test conditions

Test Condition	Direction	Average Response Time (ms)	Voltage Threshold
Indoor (quiet)	Forward	180	> 2.8V (Y-axis)
Indoor (quiet)	Left	185	< 2.2V (X-axis)
Outdoor (uneven)	Right	190	< 2.2V (X-axis)
Outdoor (uneven)	Stop	160	~2.5V (X/Y neutral)

4.1.6 Motor and Driver Performance

The motor control subsystem used the BTS7960 43A high-power motor driver module to control two 350W 24VDC motors with independent connectivity to allow the individual motors to be steered differently to each other. PWM Analysis: A set of preliminary tests applying maximum PWM (255) provided an excess speed. To be safe and easily controllable, the final evaluation setting became PWM = 200.

Directional Performance: Uniform reactions were realized with both motor drivers when subjected to similar PWM inputs and forward and backward movements were stable. The measurements of voltages on the outputs of the drivers were steady at about 23.5V.

Turning Functionality: The use of PWM on one motor and shut off on the other motor resulted in turning functionality at stationary position. Certain minor adjustments of the mechanical nature had to be made in order to rectify the minor problems relating to the chain alignment.

Stop Response: Stopping the vehicle by having all PWM channels at 0 showed that there was an instant stop and no observable drift demonstrating the responsiveness of the drivers.

Driver Synchronisation: The BTS7960 left and right modules were also synchronized well thus allowing trustworthy and synchronized movement. Operational Conditions: •

(i) Supply: 24V two 12V batteries fitted together in series. • Terrain: Straight level indoor surface.

(ii) Load capacity: 55kg of approximate payload (chassis and electronics).

(iii) Run-Time: Non-stop 2-3 minutes with cooling breaks.

Safety Considerations: 43A rating of the BTS7960 safely surpassed the motor current requirements. Short-duration tests did not notice any overheating. Enable pins used were used to provide emergency stops and controlled activation.

4.2 Mechanical Evaluation

According to a preliminary mechanical analysis of the wheelchair system, there are a number of problematic issues that influence the drivetrain performance. First of all, the wheel-to shaft connection failed: the shaft would turn without turning the wheel, indicating a slack coupling mechanism or hub connection. Also disturbing was the issue of chain tension, which turned out to be a significant concern of reliability; chain constantly derailed off of the sprocket because of misalignment or over-slack conditions, particularly in turning manoeuvres and directional changes, making movement unstable. To eliminate such shortcomings, the mechanical assembly was sent back to the workshop, where major changes were made. These were mainly changing the wheel-to-shaft coupling, the redesign of the sprocket orientation as shown in figure 4.4, and an adequate mechanism of chain tensioning. These fixes enhanced mechanical stability significantly and guaranteed the absence of stall in power delivery between the motor and the wheels.

4.3 Safety and Fail-Safe Strategy

On the target of protecting operational reliability and ensuring end-users safety, the current prototype currently uses a layered architecture with a fail-safe strategy. The joystick serves as a manual backup, providing the direct propulsion command with failure of the voice recognition system or wireless-based communication network. The arrangement proves essential where a high ambient noise level, unsteady radio channels, or a mismatched voice input is involved. The Arduino firmware has a priority control scheme which decides that in case joystick is detected, the voice commands would be overwritten. This type of prioritization ensures that the user can regain instantaneous control without the assistance of unwanted commands. Failure scenarios,

simulated widely, showed that losing the Wi-Fi connectivity and voice input did not interrupt the ongoing work in a timely manner due to the joystick adapting to the situation and reacting to new circumstances. This multi-level architecture therefore contributes to improved reliability as well as safety especially where mobility is essential in critical settings regardless of the state of a system.

4.4 User Evaluation

To assess the real-world functionality and user experience of the hybrid wheelchair system, an informal user evaluation was conducted. This phase focused on key performance indicators such as voice command accuracy, joystick responsiveness, comfort, accessibility, and overall system usability. Participants included three volunteer users, who were given simple navigation tasks such as moving forward, turning, and stopping using both voice and joystick inputs. Evaluations were observed in a controlled indoor environment with occasional background noise to simulate semi-realistic usage conditions.

Voice Command Performance: The system demonstrated a 90–92% accuracy rate for voice command recognition in quiet conditions and dropped slightly to 84–87% in environments with mild ambient noise. Commands such as "Forward" and "Stop" were most reliably interpreted, while directional commands like "Turn Left" and "Turn Right" occasionally required repetition. Users found the two-second listening window useful in reducing false positives, though a few noted the need for better voice feedback or confirmation.

Joystick Responsiveness: The joystick, acting as a backup navigation method, performed consistently well, with precise directional control and negligible latency. However, turning manoeuvres were initially reported to be too sharp. This was resolved by introducing speed

balancing during left and right turns, which users reported felt more natural and stable. One user noted improved confidence using the joystick after becoming familiar with its sensitivity range. Ease of Use and Comfort: Users rated the system as moderately easy to operate, especially after a short familiarisation period. The dual-input control system increased their sense of security, knowing that loss of voice function could be backed up by manual control. Ergonomically, the button placements and joystick location were deemed appropriate. However, users recommended a slight repositioning of the voice command indicator (LED) for better visibility. Power and Safety Feedback: The manual power button and auto-shutdown after 1 minute of inactivity were appreciated as safety measures. Users understood the visual cues provided by the LED blinks for commands but suggested adding auditory feedback in future iterations. Overall System Rating: On a scale of 1 to 5, the average overall system satisfaction rating was 4.2. The combination of offline voice recognition, manual joystick backup, and reliable motor responses contributed to the system being perceived as practical, assistive, and empowering for individuals with limited mobility. The main critique was the system's dependency on clear articulation and lack of real-time voice feedback, both of which are areas for future enhancement.

4.5 Challenges Encountered

Several technical and mechanical challenges were experienced on the project. The instability in the connection between the NodeMCU and the MIT App was quite remarkable, and this can be attributed to the fact that the mobile device uses time synchronized Wi-Fi communication. As a result, time drift resulted in the creation of lapses sometimes. The extraction of MAC addresses was an issue in the context of the Kodular and the programmatic connection to such Bluetooth modules like the HC-06 was restricted. Various tests were required to come up with a stable

communication technique. Noise in the environment was becoming a problem to voice recognition when it comes to reliability; it was hard to recognize the commands issued in a noisy atmosphere with other people talking in the background. Mechanically, the system chain and sprocket had become poorly aligned such that the shaft was rotating but the wheels were not moving in a sign of improper coupling or insufficient tension. Adjustments gained in the laboratory were necessary to provide dependable movement.

4.5 SUMMARY OF RESULTS

Table 4. 3: Summary of Results

Test Parameter	Observation/Result
Voice command accuracy	Achieved 85–90% recognition accuracy in quiet indoor conditions. Accuracy dropped to 60–70% in noisy environments due to background interference, despite using offline speech recognition.
Wi-Fi control delay	Average delay between command from MIT App and execution via NodeMCU–Arduino communication was measured to be under 1 second. This response time remained consistent under a stable network
Joystick response	The joystick offered immediate, real-time control with smooth transitions between directions. Analog signal readings from the joystick ranged between 0V to 5V, accurately mapped for directional movement.
Motor driver effectiveness	The BTS7960 driver modules delivered stable control with consistent PWM output. Full speed achieved at PWM = 255; tested safer operation at PWM = 200. Drivers responded promptly to both voice and joystick inputs.
Mechanical Alignment	Initially faced misalignment where the shaft rotated without driving the wheels. After corrections at the mechanical lab (re-securing sprocket-to-wheel connection and adjusting tension), system alignment improved to a fair and functional level.

4.6 Discussions

In the current research, an innovative voice-controlled intelligent wheelchair system was tested, one which uses a combination of verbal commands and manual joystick inputs typically called in combination as a hybrid assistive control. The authors suggest that the two modalities of input give it more accessibility and usability, especially in scenarios where using any one mode of input becomes unreliable, or inappropriate, e.g. environmental noise making voice recognition unreliable, or speech impairments making clear speech transmission a no-go. The main conclusion of the research is related to the addition of an offline-enabled MIT application that removed the need to always be on the internet. Under quiet conditions, offline recognition was able to perform sufficiently well but in dynamic or noisy conditions, there was a serious decline in voice command accuracy.

The implementation of remote command transmission using Wi-Fi and NodeMCU made possible a satisfactory connection infrastructure of the IoT-enabled scooter, nevertheless, some challenges were met that accentuate the trade-offs within the wider Bluetooth and Wi-Fi ecosystem. Periodic obstacles were first-pairing instabilities, incompatible IP-addresses and high latency during network stresses and this propounded the fragility of the relationship of ease of use but poor interoperability of Bluetooth against versatility (compensated with latency and system complexity) of Wi-Fi. Mechanically, a problem existed in coupling the motor shaft to the rear wheel that caused a slipping of the wheel and an unbalanced mode of propulsion; motor and wheel were not well coupled. These mechanical shortcomings did not allow a prerequisite functional reliability assessment in all command conditions and stressed the need of stronger mechanical connection between the drive train and the wheel assembly.

The joystick was used as an important fail-safe system, because it was possible to always exercise manual control, even when the voice commands failed or communication was shut down temporarily. Its responsiveness, in real time, was invaluable to the reliability of the whole system. Despite the modularity of the system, which made it more flexible and potentially scalable in the future, due to the integration of the MIT App, NodeMCU, Arduino, BTS7960 and joystick, the challenges to synchronise these parts in the continuous workflow were also identified.

CHAPTER FIVE

CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusions

This initiative has created a smart wheelchair with the capability of having a hybrid control which is a combination of an offline voice recognition tenderness and manual Joystick control which will encourage better mobility and safety. The architecture, which is depicted in Fig. 1, includes a mobile application developed using the MIT App Inventor tool, a NodeMCU (ESP8266) module to act as an access point to connect to the Wi-Fi network, and an Arduino Uno board acting as a central command. Such setup facilitates the possibility of a dual mode, appropriate to users with a wide range of physical skill as well as level of speech. The most important results of this development are the following:

i. The offline voice-control agent works with a comparatively low level of background noise, as in most laboratory tests, its average recognition accuracy is more than 80 % as shown in Table 4.1.

ii. The manual joystick acts as a reliable fail-safe system, especially in the case of communications failure or recognition error; this further has advantages of enhancing the system strength.

iii. Though early Bluetooth connectivity issues were experienced, the later change of architecture to Wi-Fi enhanced the stability of the system when appropriately set-up.

iv. Wheel-shaft disengagement in manual-shift operations, and irregular wheel-shaft chain tension were significant in terms of operations in prototyping but could be fixed by careful tuning in the laboratory environment.

To summarise, the wheelchair system under consideration here can be described as a scalable, cost-efficient, semi-autonomous mobility system, an idea which is particularly useful within low-resource settings.

5.2 Recommendations

In order to improve the assistive mobility systems in the future, several prospective recommendations are introduced to reinforce the effectiveness, independence, and user-friendliness of voice-controlled wheelchairs.

To begin with, the next generation of systems should learn to utilize the advantages of AI-powered microcontrollers e.g., the ESP32-S3 or other edge-AI platforms that feature the ability to provide embedded neural networks to process and respond to voice requests and make real-time decisions without referring to external mobile devices. Such enhancement would enhance autonomy and responsiveness.

As well, advanced sensor fusion can enhance the navigation accuracy and safety. Future versions may include LIDAR, inertial measurement units (IMU), and computer see hats to supplement and improve on the rudimentary ultrasonic or IR sensors. Such sensors together could be used to detect the terrain, avoid obstructions, and actively plan the path especially in dynamic and / or unpredictable circumstances.

Technologically, the system may become adaptive and modular chassis design mechanical-wise. The chassis might be fitted with self-configuring chain or belt powered drive systems with diagnostic, and feedback-controlled actuators which adapt to individual motion profiles or terrain types, improving ride quality, as well as reliability.

Moreover, voice control system itself might be even made smarter by going beyond a simple keyword-driven functionality to a complete natural language understanding (NLU). The wheelchair could interact in more subtle ways with its user by using tiny machine learning (tinyML) models running on edge devices.

A combination of a modern wheelchair system with cloud connection brings with it a legion of new possibilities, such as real-time diagnostics, automatic firmware updates, and the competence to share aggregated, anonymous data between multiple devices. By using federated learning, such devices are able to hone their performance collectively, and at the same time do not harm user privacy. In addition, with the inclusion of newer forms of Human-machine interaction modalities, e.g. EMG (electromyography) or EEG (electroencephalography) signals, further control mechanisms can be given to the users expecting severely limited motor abilities. These inventions, together with the use of predictive analytics based on extended field tests, will not only improve the performance of the system, but will also foster the design of the next products that will be more adaptive and safer as well as adaptive to the heterogeneous combination of users.

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APPENDICES A – BILL OF MATERIAL

Material Description	Qty	Amount (Naira)	Total Amount (Naira)
Arduino UNO Microcontroller	1	13,000	13,000
Node MCU ESP8266	1	8,000	8,000
Gear Motors	2	115,000	230,000
Wheelchair frame	1	150,000	150,000
Chain and Sprocket Integration	2	60,000	120,000
Cables and Connectors		5,000	5,000
BTS7960 Motor Drivers	2	8,000	16,000
DC-DC Buck Converter	1	4,000	4,000
Emergency Push-Button	1	1,000	1,000
Joystick Controller	1	800	800
Power LED	1	100	100
Miscellaneous		50,000	50,000
TOTAL			**597,900**

APPENDICES B – PROJECT SOURCE CODES

All the codes related to this project can be found in the drive below;

https://drive.google.com/drive/folders/1LwFpwjkoIshTZC4paYUe4R1mfeWB47va?usp=sharing

NodeMCU and Arduino codes should be ran using the Arduino IDE while the Wheelchair App .aia file should be imported into the MIT app Inventor.

APPENDICES C – DESIGN CALCULATIONS

All design calculations can be found in the link below;

https://drive.google.com/drive/folders/1LwFpwjkoIshTZC4paYUe4R1mfeWB47va?usp=sharing